Chapter 6 Compemsator Design

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§ 1 Introduction

1.1 Necessities of Compensation

A system may be unsatisfactory in

- **stability**
- **speed of response**
- **steady-state error**

1.2 Performance Specifications

- **Accuracy:** K_p, K_v, K_a
- **Relative stability:** K_g, g
- **Speed of response:** w_r, M_r

($M_r = 1/(2z\sqrt{1-z^2})$)

NB:

- **High performance means high cost.**
- **Required performance specifications may be contradictory to each other.**
- **Practical specifications**

  In time domain: $t_r, M_p(s)$, t_s

  In frequency domain: g, K_g, M_r, w_B

Desired values usually adopted:

$45° \le g \le 60°$, $K_g \ge 10$ **dB,** $1.0 < M_r < 1.4$,

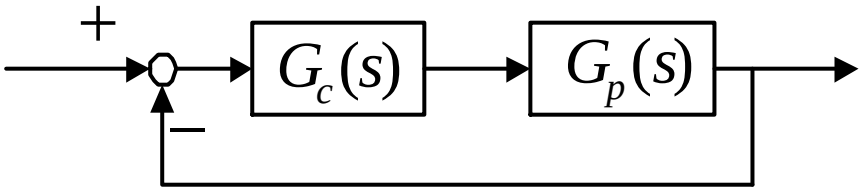
$0.4 < z < 0.7$, $M_p < 0.25$.

For a typical 2nd order system

z	g	M_p	M_r
0.4	**44°**	**0.254**	**1.364**
0.7	**60°**	**0.046**	**1.002**

1.3 System Configurations

1. Cascade Compensation



¨ **Features:**

à **Simple**

à **Additional amplifier**

¨ **Compensator types:**

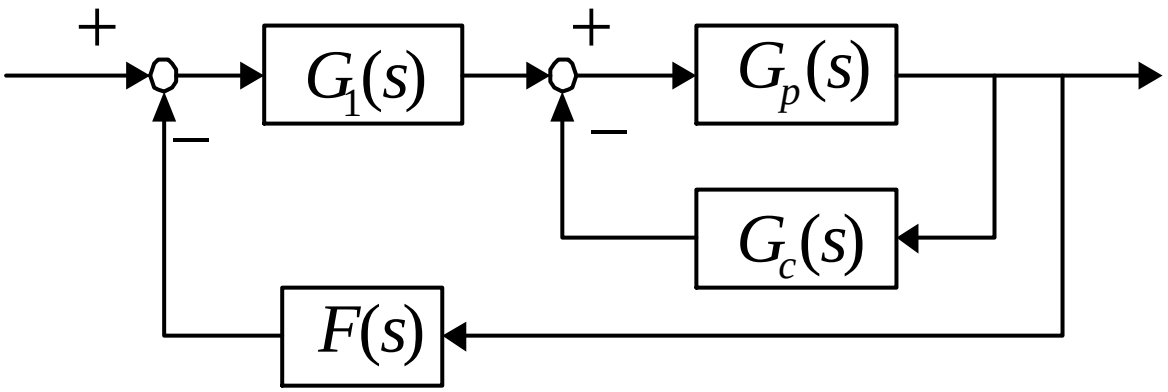
à **Lag compensator: to raise steady state accuracy**

à **Lead compensator: to improve transient response**

à **Lag-lead compensator: to raise steady state accuracy and response speed**

à **Bridged-T compensator (Notch filter):**

2. Feedback Compensation



¨ **Features:**

à **Complicated in analysis and design;**

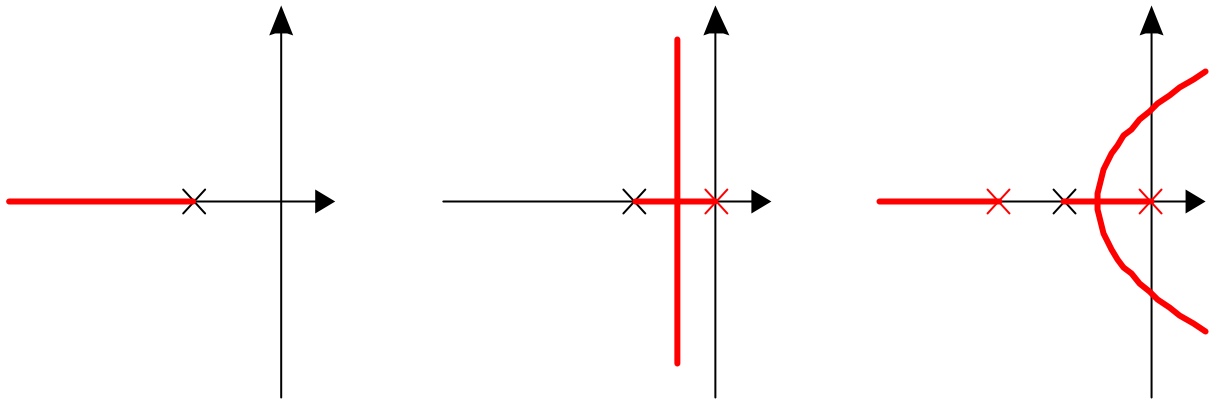
à **Special results may be obtained by using simple compensator.**

1.4 Design methods

1. Root Locus Approach

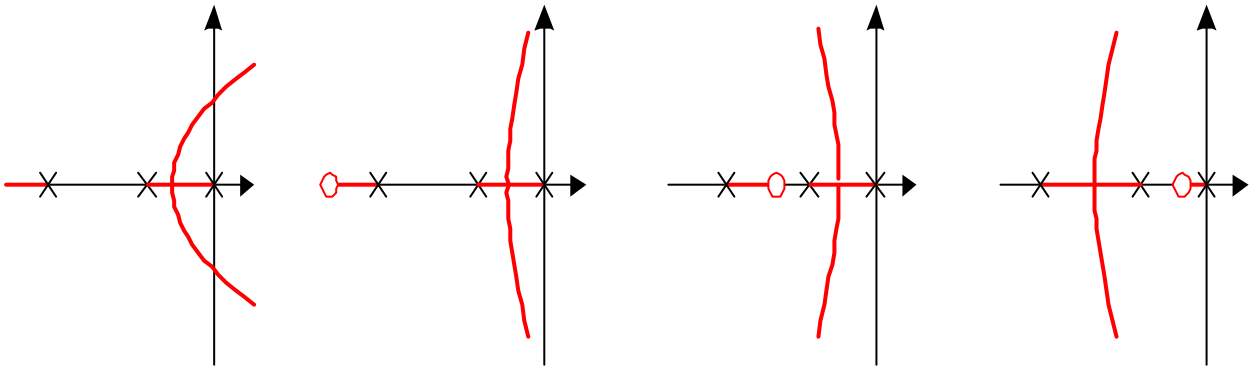
(1) Poles move the locus to the right

Þ lower stability of the system



(2) Zeros move the locus to the left

Þ fast response and higher stability



(3) The gain is adjusted such that the dominant poles correspond to the higher gain, and consequently e_{ss} will decrease.

(4) The common specifications used are z, w_n and K_v

2. Frequency Response Approach

(1) The common specifications used are g , K_g, M_r and w_r.

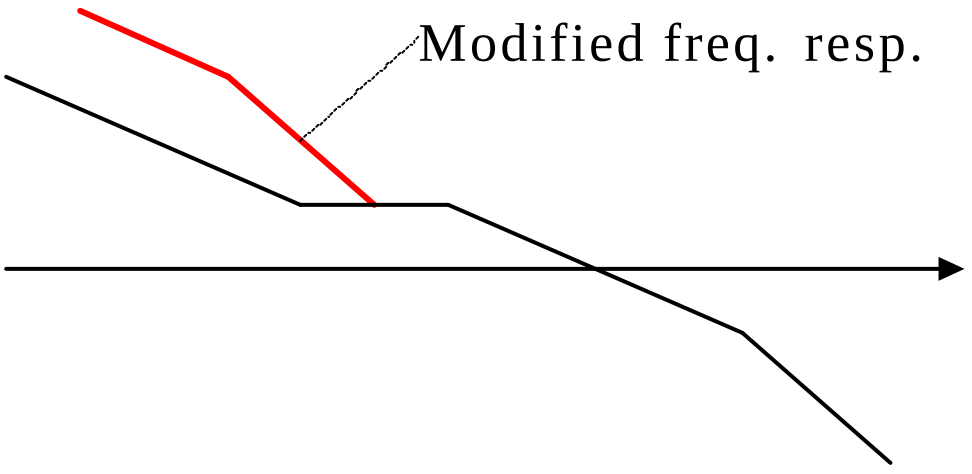
- $M_r = \dfrac{1}{2z\sqrt{1-z^2}}$ **represents the damping ratio**
- $w_r = w_n\sqrt{1-2z^2}$ **represents the fastness**

(2) Relations between the frequency response and system performance:

- **Lower frequency – steady state error**
- **Middle frequency – relative stability**
- **Higher frequency – speed of response**

(3) Unsatisfactory performances

(i) Stable but e_{ss} is too big

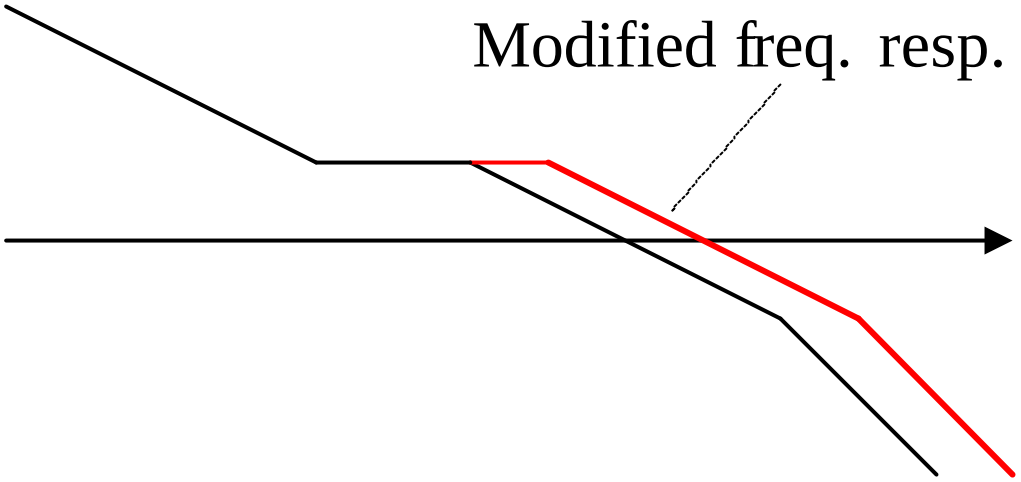


- **The middle frequency response**

ensures the stability.

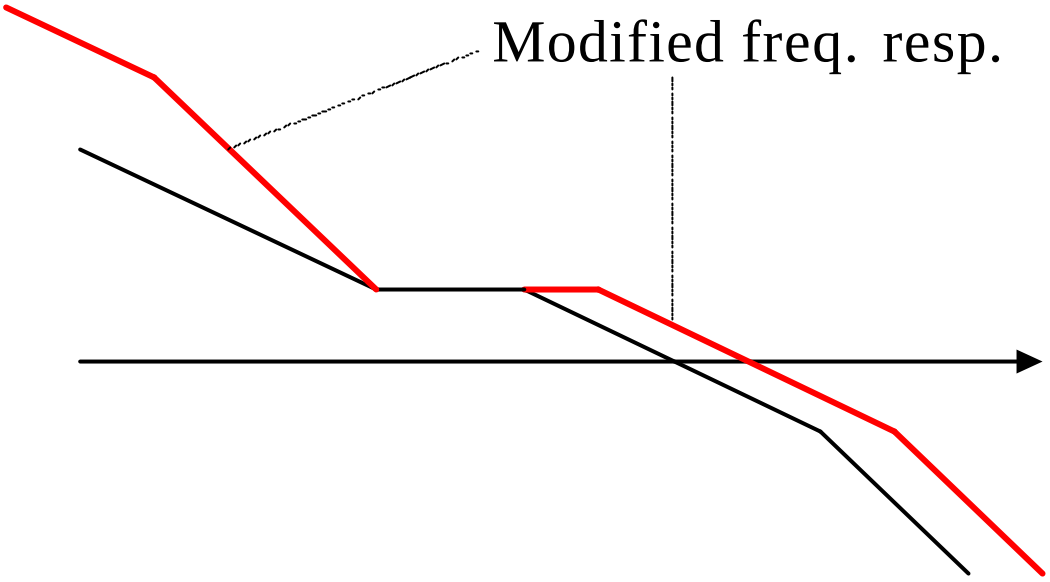
- **The magnitude of the response at lower frequency is too low.**

(ii) Stable but time response is too slow



- **The middle frequency response ensures the stability.**
- **But w_c is not high enough.**

(iii) Stable but e_{ss} is too big and time response is too slow



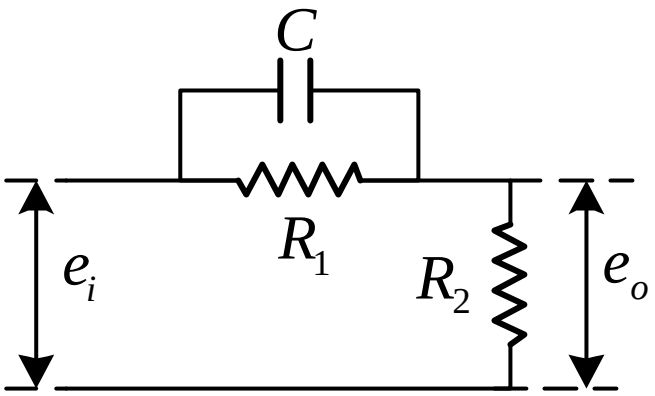
- **The middle frequency response ensures the stability.**
- **But the magnitude of the response at lower frequency is too low.**
- **And w_c is not high enough.**

(iv) Unstable for any *K*.

- **The OL frequency response is not appropriate.**
- **The most parts of the frequency response should be modified.**

§ 2 Lead Compensation

2.1 Phase Lead Network



$$\frac{E_o(s)}{E_i(s)} = \frac{R_2}{R_2 + \dfrac{R_1 \dfrac{1}{Cs}}{R_1 + \dfrac{1}{Cs}}} = \frac{R_2}{R_1 + R_2} \cdot \frac{R_1 Cs + 1}{\dfrac{R_1 R_2}{R_1 + R_2} Cs + 1}$$

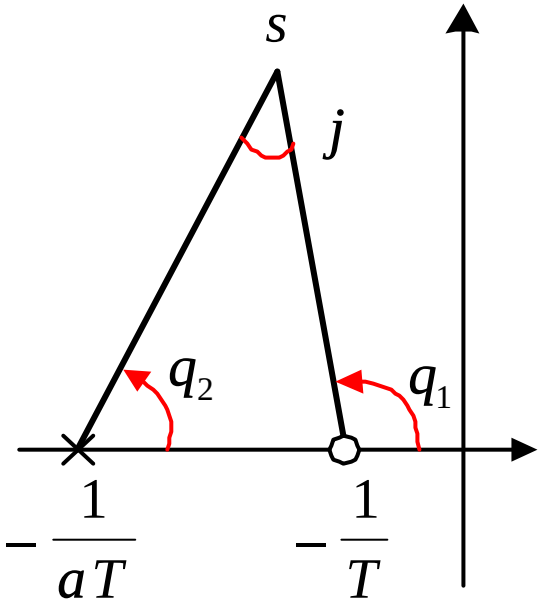
Let $R_1 C = T$, $\dfrac{R_2}{R_1 + R_2} = a < 1$, **then**

$$\therefore \quad \frac{E_o(s)}{E_i(s)} = a \frac{Ts + 1}{aTs + 1} = \frac{s + \dfrac{1}{T}}{s + \dfrac{1}{aT}}$$

2.2 Characteristics

$$G_c(s) = a\,\frac{Ts+1}{aTs+1} = \frac{s+\dfrac{1}{T}}{s+\dfrac{1}{aT}}$$

1. Pole-Zero allocation

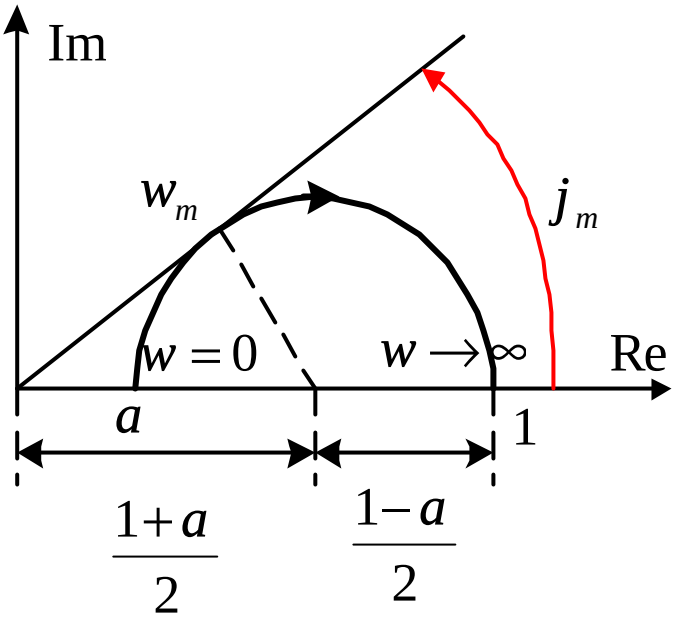


- For any test point s, $w > 0$

  leading angle $j = q_1 - q_2 > 0$.

- Moves the root locus to the left.

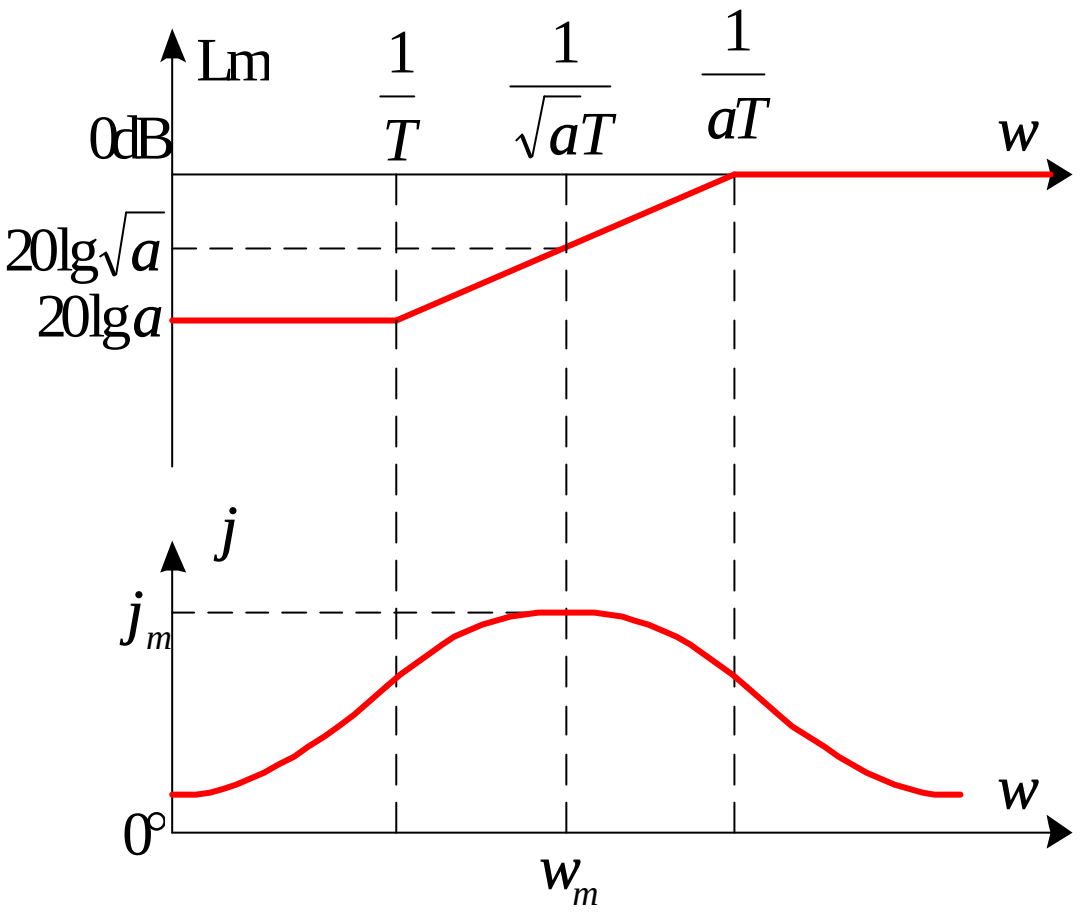
2. Polar plot



$$\sin j_m = \frac{\frac{1-a}{2}}{\frac{1+a}{2}} = \frac{1-a}{1+a}, \quad a = \frac{1-\sin j_m}{1+\sin j_m}$$

$$\therefore \quad j_m = \arcsin\frac{1-a}{1+a} = \arctan\frac{1-a}{2\sqrt{a}}$$

3. Bode diagrams



- **Behaves like a high pass filter.**
- $w_m = \frac{1}{\sqrt{a}T}$ **is the geometrical centre of** $\frac{1}{T}$ **and** $\frac{1}{aT}$**.**
- **When** $a = 0.1$, $j_m = 54.9°$**.**
- $\mathrm{Lm}G(\mathrm{j}0) = 20\lg a < 0\ \mathrm{dB}$**.**
  - Þ **An additional gain is required for steady-state accuracy.**

2.3 Phase Lead Compensation by Using Root Locus Method

1. Ideal Derivative Compensation

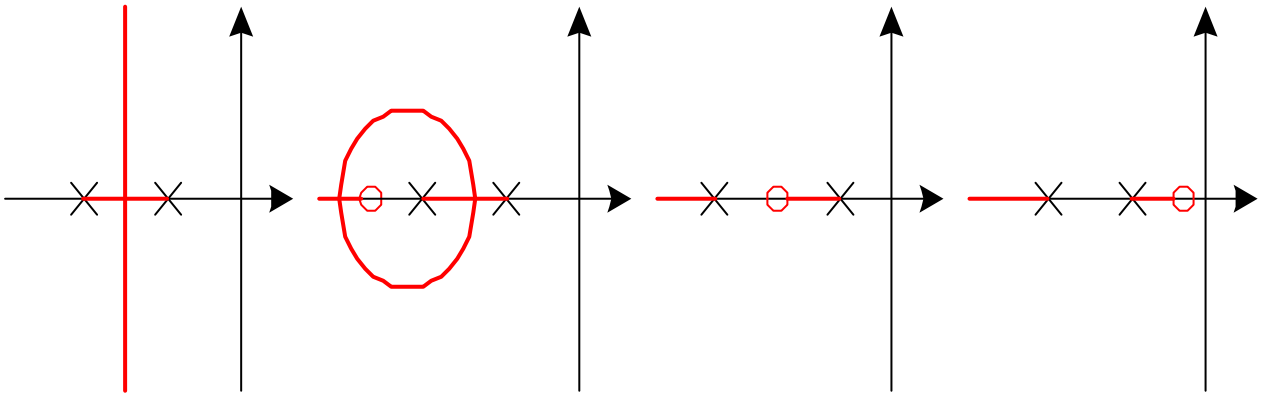
¨ **Slow response (instability) implies that the CL poles are too near to the imag. axis (in the Right Half Plane).**

¨ **Remedy:**

Adding a single zero ($G_c(s) = 1 + T_d s$)

Þ **moves root locus to the left**

achieves the higher stability



¨ **Shortcomings:**

à **Difficult to realize** $G_c(s) = 1 + T_d s$**;**

à **The noise is amplified,**

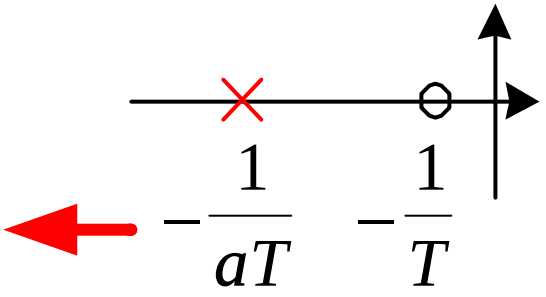
esp. the higher freq. noise.

2. Phase Lead Compensator

$$G_c(s) = K_c \frac{s + \frac{1}{T}}{s + \frac{1}{aT}} = K_c \frac{s + z_c}{s + p_c}, \quad a < 1$$

¨ K_c **compensates the lower frequency gain.**

¨ **It is desired that** $\frac{1}{aT}$ **be large**

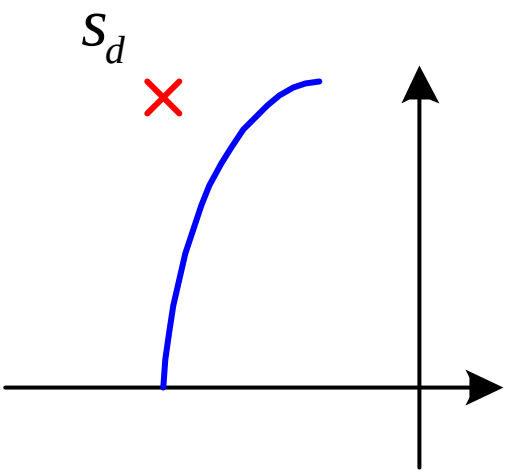


à $-p_c = -\frac{1}{aT}$ **be far away from the origin.**

à **Effect of the pole of the compensator on the root locus can be neglected**

à **But a pole far away to the left causes small** a **and consequently the difficulty in realization.**

3. Required Phase Lead Angle for a Dominant Pole



- **Expected dominant pole:**

$$s_d = -zw_n + jw_n\sqrt{1-z^2}$$

(not on the original root locus)

- **The phase angle condition:**

$$\arg\left[G_p(s_d)G_c(s_d)\right] = \arg\left[G_p(s_d)\right] + \arg\left[G_c(s_d)\right]$$
$$= \pm 180^\circ(2k+1)$$

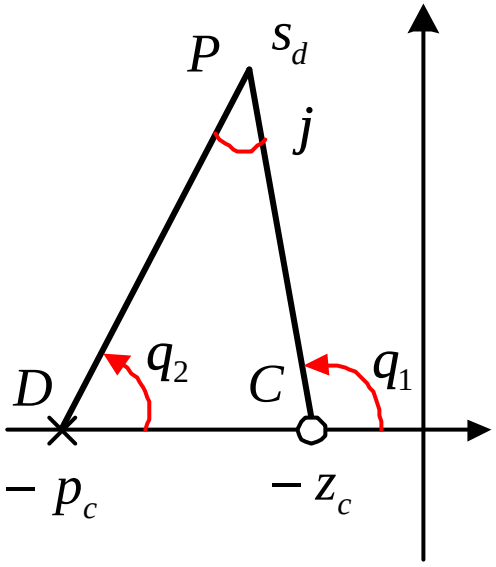
- **Phase lead angle provided by Compensator:**

$$j = \arg\left[G_c(s_d)\right]$$
$$= \pm 180^\circ(2k+1) - \arg\left[G_p(s_d)\right]$$
$$\arg\left[G_c(s_d)\right] = \arg(s_d + z_c) - \arg(s_d + p_c)$$

$$= q_1 - q_2$$

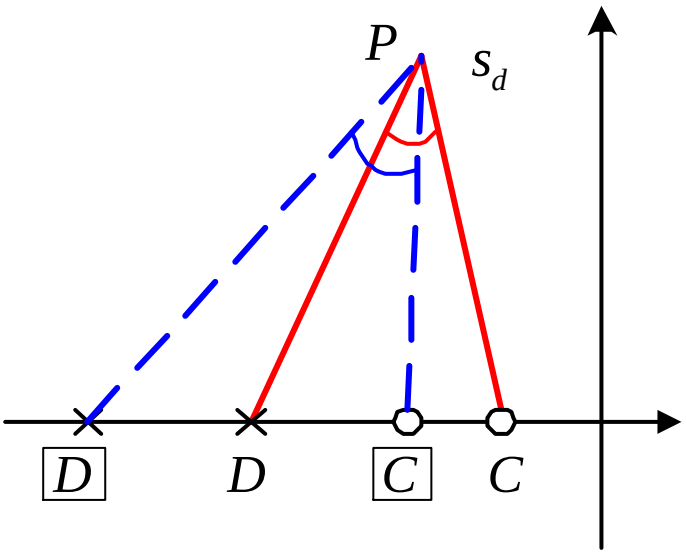
$$j = q_1 - q_2$$

¨ **Proper choices of p_c and z_c assure the right phase lead angle.**



4. Selection of p_c and z_c

(1) Infinite number of solutions for a required phase lead angle j



- **Different pairs (p_c / z_c) correspond to different a and different open loop gains.**

(2) Relation between $p_c(z_c)$ and OL gain

- **Magnitude condition**

$$\left|G_p(s_d)G_c(s_d)\right| = \left|G_p(s_d)\right| K_c \left|\frac{s_d + z_c}{s_d + p_c}\right| = 1$$

$$K_c = \left|\frac{1}{G_p(s_d)}\right| \left|\frac{s_d + p_c}{s_d + z_c}\right| = \left|\frac{1}{G_p(s_d)}\right| \left|\frac{\overline{DP}}{\overline{CP}}\right|$$

- **Before compensation**

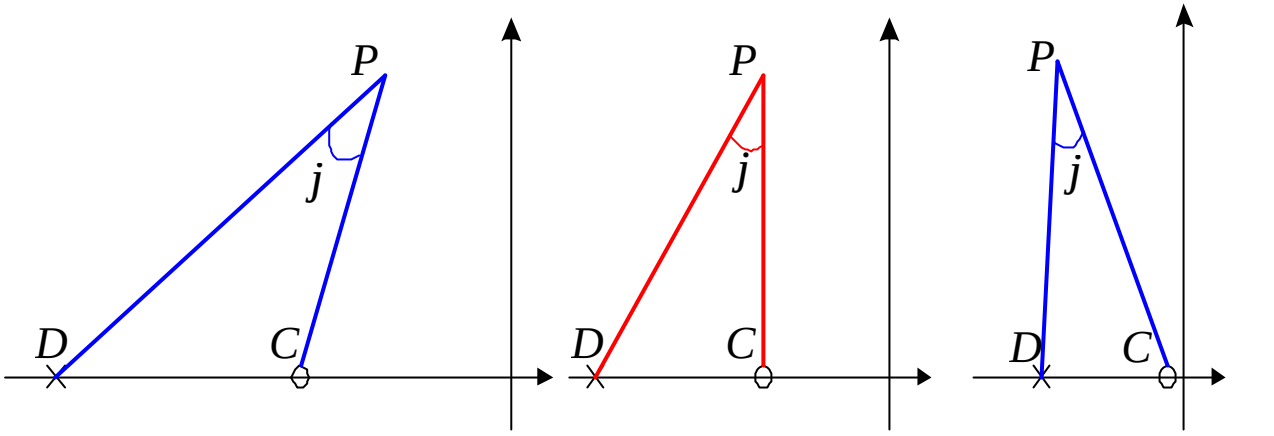
$$\lim_{s\to 0} sG_p(s) = K_v$$

- **After the compensation**

$$K_v' = \lim_{s\to 0} sG_p(s)G_c(s)$$

$$= K_v K_c \frac{z_c}{p_c} = K_v K_c a$$

¨ **3 different cases:**

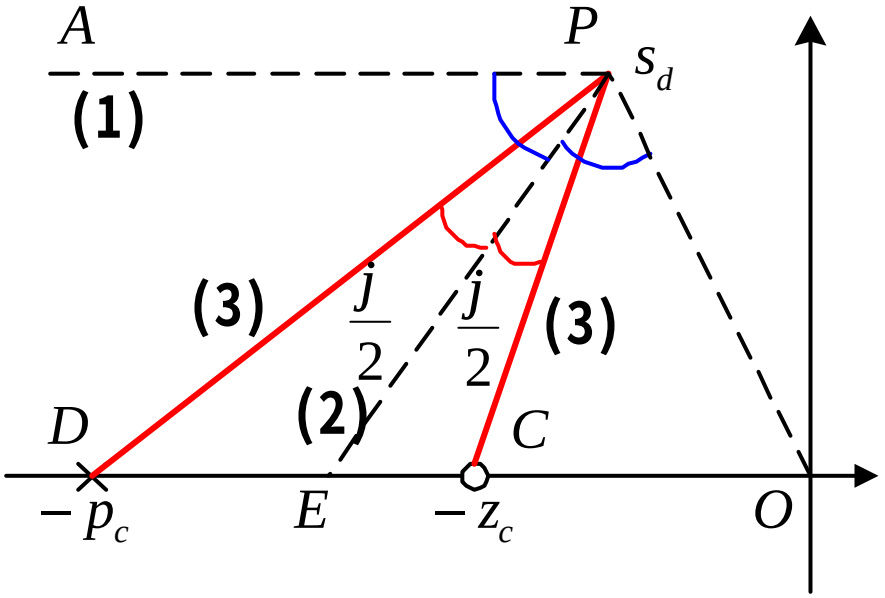


$a\downarrow, K_c\uparrow\uparrow, K_v'\uparrow$ $\qquad$ $a\downarrow\downarrow, K_c\downarrow, K_v'\downarrow\downarrow$

à **When** $p_c(z_c)$ **moves to the left**

Þ K_c - , K_v' -

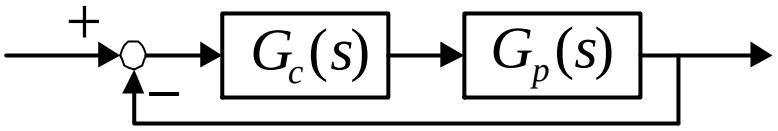
(3) Obtaining $p_c(z_c)$ **with the largest** a



$$K_c = \left|\frac{\overline{PD}}{\overline{PC}}\right| \cdot \frac{1}{\left|G_p(s_d)\right|}, \quad a = \frac{z_c}{p_c}, \quad K_v' = K_v K_c a$$

5. Example

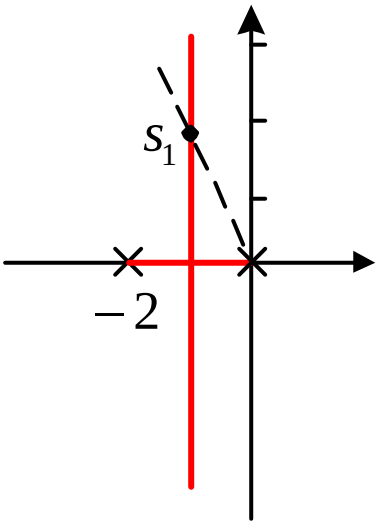
Example 2.1 Given $G_p(s) = \dfrac{1}{s(s+2)}$.



Design a phase lead compensator such that the desired CL poles have $w_n = 4$ rad/s **and** $z = 0.5$.

Solution:

(i) Uncompensated system analysis



$$G_{CL}(s) = \frac{4}{s^2 + 2s + 4}$$

$$s_{1,2} = -1 \pm \mathrm{j}\sqrt{3}$$

$$w_n = 2 \text{ rad/s}$$

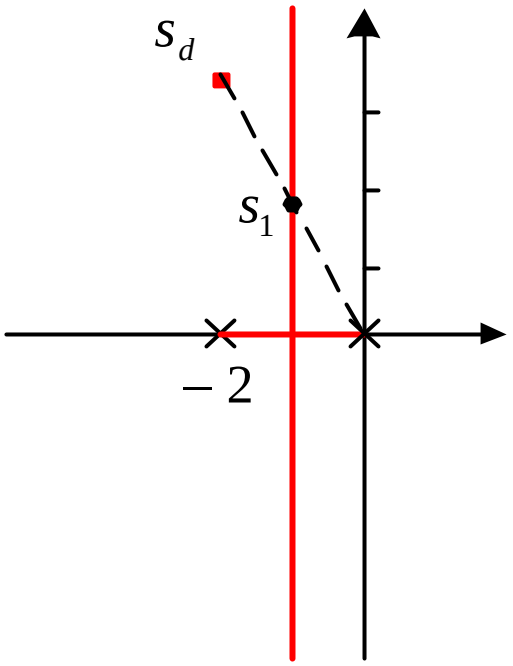
$$z = 0.5$$

$$K_v = 2\,\mathrm{s}^{-1}$$

(ii) Desired CL dominant poles

$$s_d = -z \pm \mathrm{j}w_n\sqrt{1-z^2} = -2 \pm \mathrm{j}2\sqrt{3}$$

- **CL poles are not on the desired places**
- **Adjusting *K* only cannot obtain the right CL poles**
- **s_d is on the left to the original root locus**



ß

Phase lead comp. is required

(iii) Calculation of the required phase lead angle

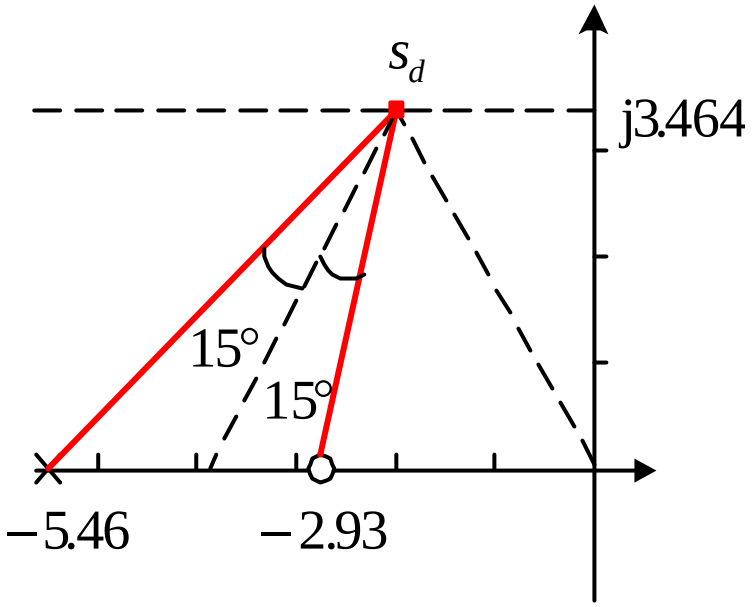
$$G_c(s) = K_c \frac{s + \frac{1}{T}}{s + \frac{1}{aT}} = K_c \frac{s + z_c}{s + p_c}, \quad a < 1$$

$$\arg\left[G_p(-2 + j2\sqrt{3})\right]$$

$$= \arg\left[\frac{4}{(-2 + j2\sqrt{3})(j2\sqrt{3})}\right] = -210°$$

$$j = -180° - (-210°) = 30°$$

(iv) Determination of p_c and z_c (a and T)



By bisector method

$$-p_c = -5.46, \quad -z_c = -2.93, \quad a = 0.5366$$

$$G_c(s) = K_c \frac{s + 2.93}{s + 5.46}$$

(v) Determine OL gain

$$G_p(s)G_c(s) = \frac{4K_c(s + 2.93)}{s(s + 2)(s + 5.46)}$$

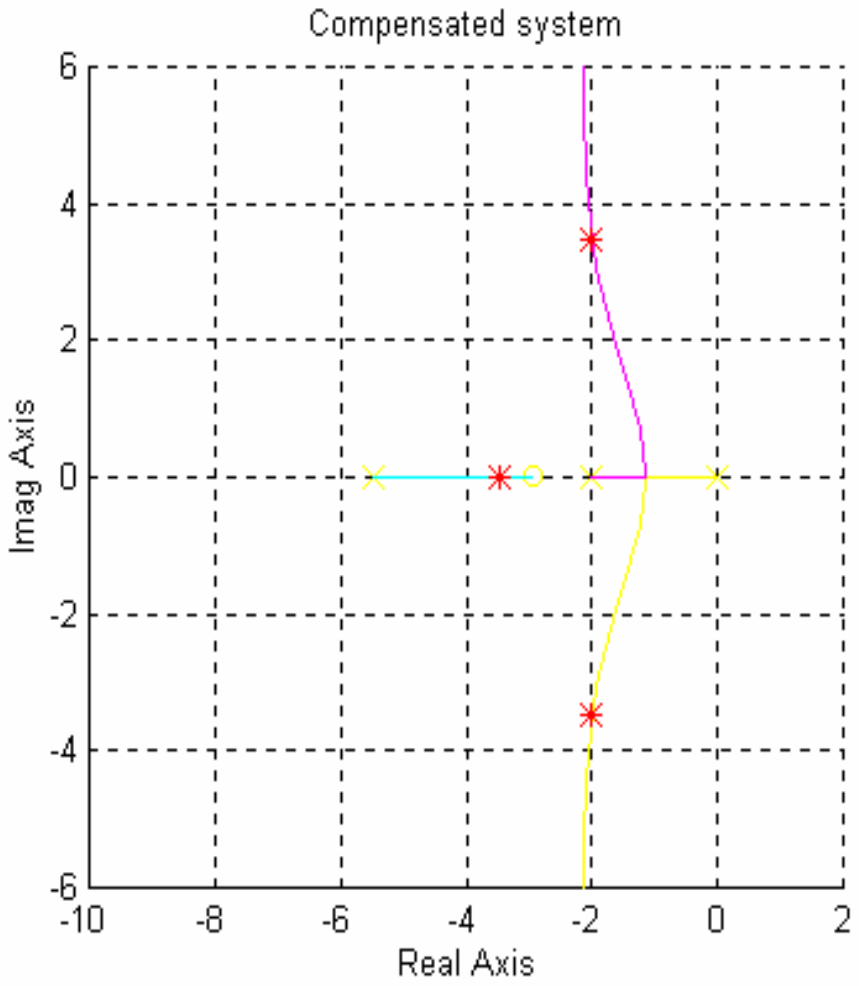
$$\left|G_p(s_d)G_c(s_d)\right|$$

$$= \left|\frac{4K_c(s + 2.93)}{s(s + 2)(s + 5.46)}\right|_{s_d = -2 + j2\sqrt{3}} = \frac{4K_c}{18.91}$$

$$K_c = \frac{18.91}{4} = 4.728$$

(vi) Check of the compensated system

$$K_v' = \lim_{s \to 0} sG_p(s)G_c(s) = \frac{18.91 \times 2.93}{2 \times 5.46} = 5.074 \text{ s}^{-1}$$



- **CLoop poles:**

$$s_{1,2} = -2 \pm j2\sqrt{3}, \quad s_3 = -3.46$$

- $s_3 = -3.46$ **is near to** $z_c = -2.93$
- **The resulted closed loop system meets**

the prespecified requirements.

6. Procedure (for reference only)

(i) Select desired dominant poles from z and w_n

(ii) Calculate the required phase lead angle j from the dominant poles

(iii) Calculate p_c and z_c from required j and prespecified K_v

s_d, j Þ **infinite solution**

s_d, j, e_{ss} Þ **unique solution**

(iv) Determine the open loop gain

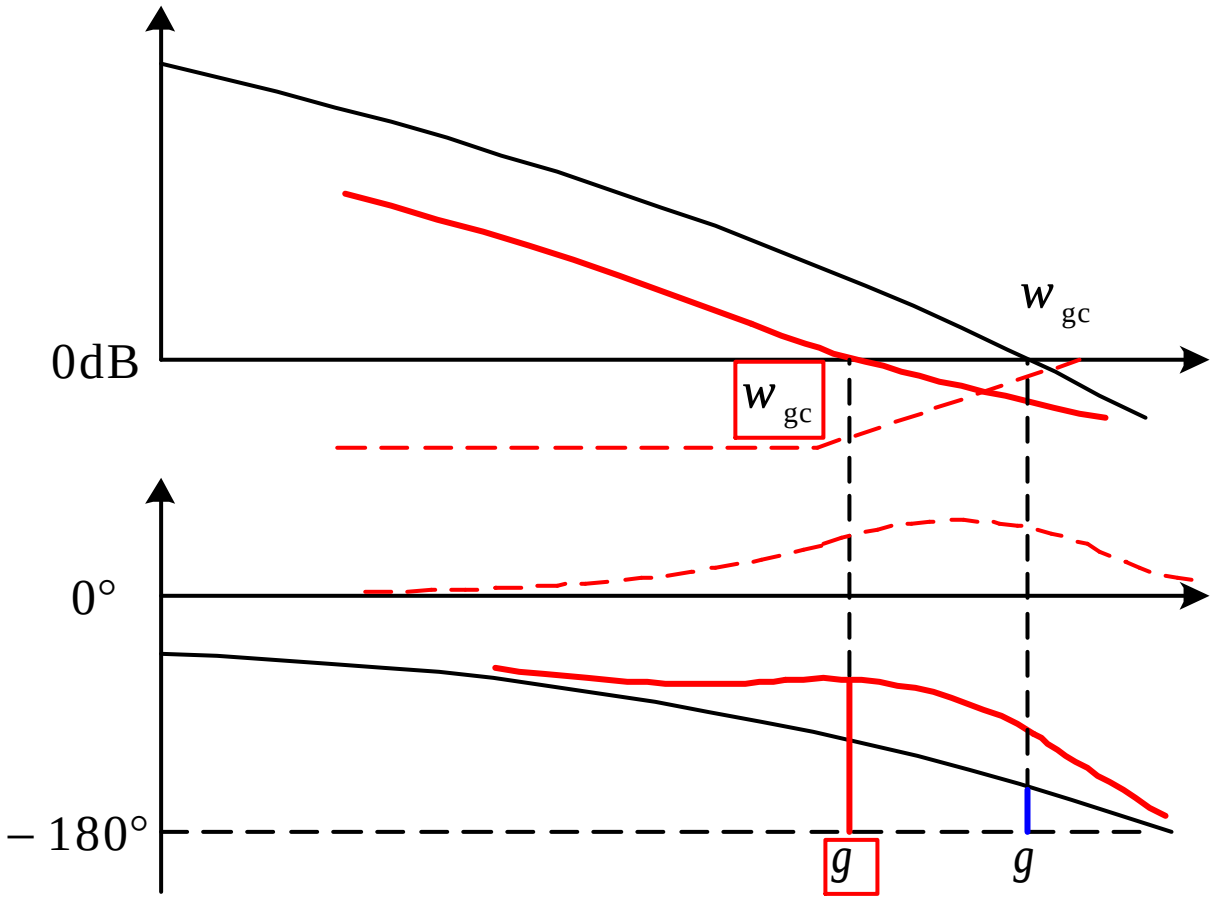
(v) System performance check and adjustment

2.4 Phase Lead Compensation by Using Bode Diagrams

1. Improvements of the performance seen from the Bode diagrams

(1) Relative stability

$$G_c(s) = a\frac{1+Ts}{1+aTs} = \frac{s+\frac{1}{T}}{s+\frac{1}{aT}}, \quad a<1$$



¨ **Lead compensation with** $K_c = 1$

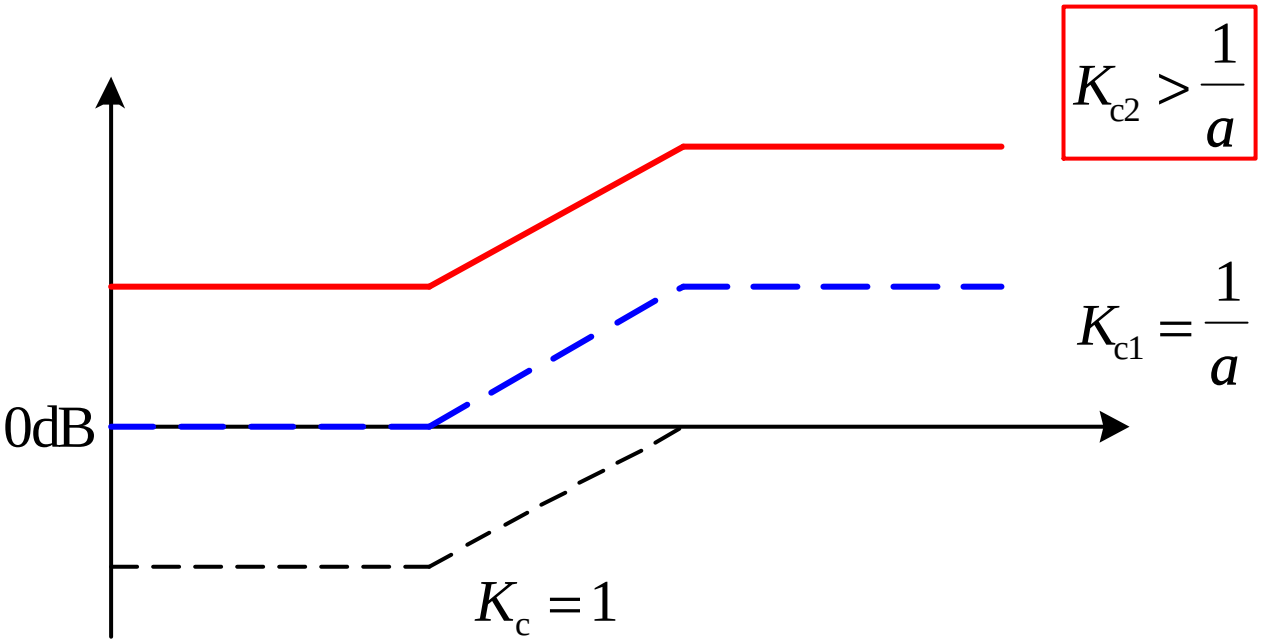
Þ w_{gc} ¯, g -

(2) Steady-state error, speed of response

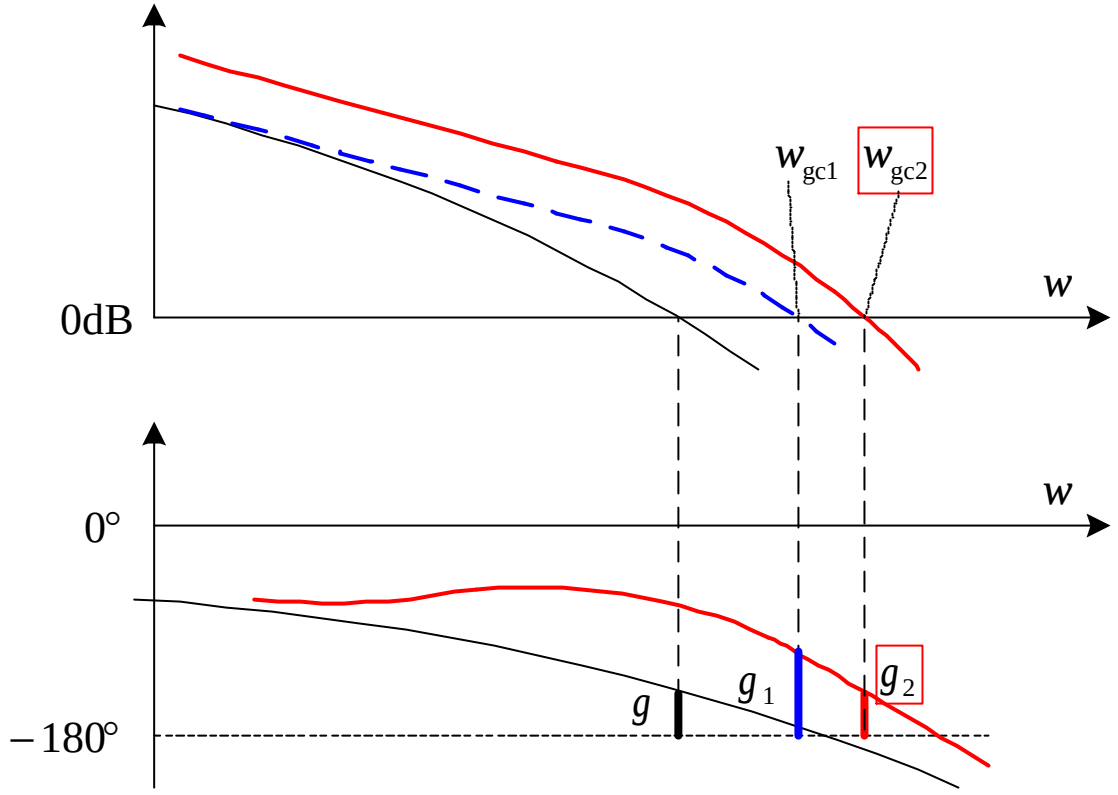
¨ **For a plant with reasonable** g

Select two compensators:

$$G_c(s) = K_c a \frac{1+Ts}{1+aTs}, \quad K_{c1} = \frac{1}{a}, \quad K_{c2} > \frac{1}{a}$$



¨ **After compensation**



à $K_{c1} = \frac{1}{a}$, w_{gc1} - -, g_1 - -, K_v **unchanged.**

à $K_{c2} > \frac{1}{a}$, w_{gc2} - -, $g_2 \approx g$ **or** g_2 -, K_v -.

2. Example

Example 2.2 $G_p(s) = \dfrac{K}{s(1+0.1s)(1+0.01s)}$

Design cascade compensator

Compensated system:

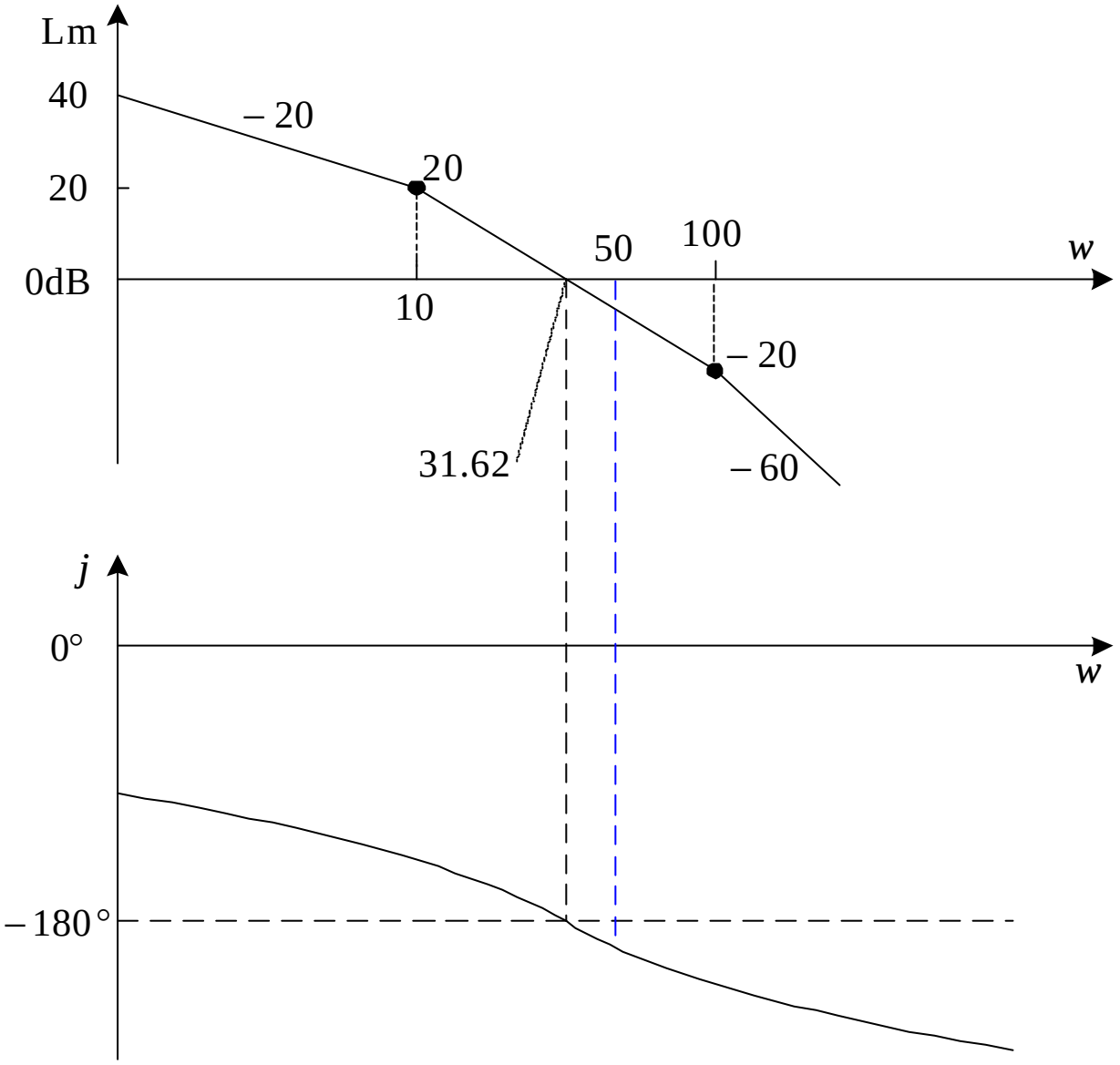
$g \geq 30°$**,** $w_c \geq 45$ **rad/s,** $K_v \geq 100\ \text{s}^{-1}$**.**

Solution:

(i) Determine desired OL gain *K*.

$$K_v = \lim_{s\to 0} sG_p(s) = K, \quad K = 100$$

(ii) Gain-compensated system analysis



à $w_{gc} = \sqrt{10 \times 100} = 31.62$ **rad/s,** $g = 0°$

à $w_{pc} = 31.62$ **rad/s,** $K_g = 0.8277$ **dB**

(iii) Required phase lead angle

¨ **Let** $w_c = 50 > 45$ **rad/s, then**

$$\arg G_p(\mathrm{j}50) = -90° - \arctan 5 - \arctan 0.5$$

$$= -90° - 78.69° - 26.57° = -195.26°$$

$$\gamma > 30^\circ \quad \text{Þ} \quad \phi = 45^\circ$$

à $\phi_m = \phi + (5 \text{ to } 10)^\circ = 55^\circ$

(iv) Calculate α

$$\alpha = \frac{1 - \sin\phi_m}{1 + \sin\phi_m} = 0.1$$

(v) Calculate T

¨ **Take** $\omega_m = 50$ rad/s**, then**

$$T = \frac{1}{\sqrt{\alpha}\omega_m} = 0.06325, \quad \frac{1}{T} = 15.8115,$$

$$\alpha T = 0.006325, \quad \frac{1}{\alpha T} = 158.1153$$

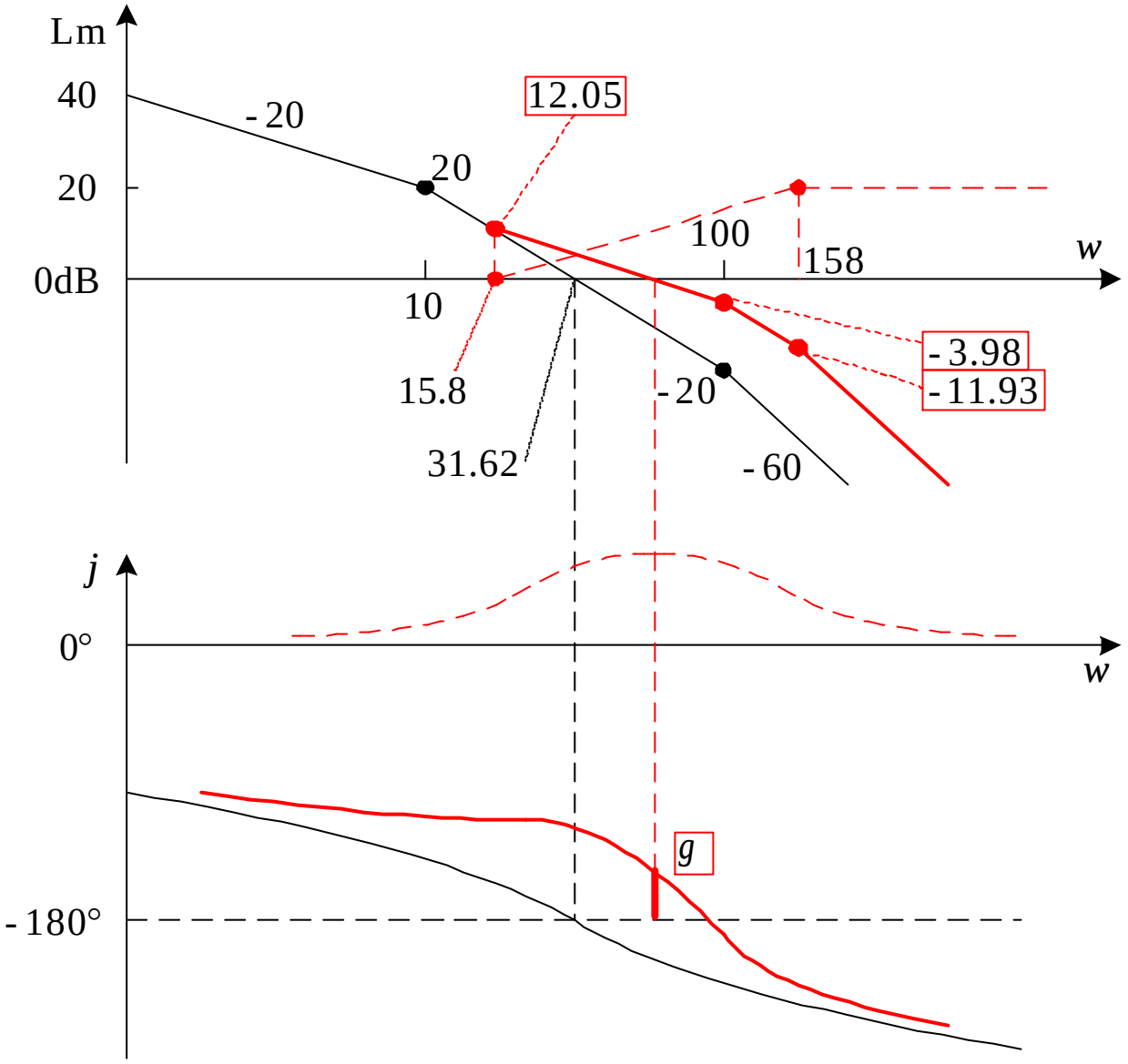
(vi) Compensator

$$G_c(s) = \frac{1 + 0.06325s}{1 + 0.006325s} = 10\frac{s + 15.8}{s + 158}$$

¨ **Keep the low frequency response unchanged for retaining** $K_v = 100 \text{ s}^{-1}$

(vii) Final check

$$G_p(s)G_c(s) = \frac{100(1 + 0.06325s)}{s(1 + 0.1s)(1 + 0.01s)(1 + 0.006325s)}$$



- **By asymptotes:**

$$w_{gc} = 63.3 \text{ rad/s}, \quad g = 30.8°$$

$$K_v = 100 \text{s}^{-1}$$

- **By Actual curves:**

$$w_{gc} = 54 \text{ rad/s}, \quad g = 36.94°$$

$$w_{pc} = 119.65 \text{ rad/s}, \quad K_g = 11.31 \text{ dB}$$

3. Procedures (for reference only)

(1) Determine the desired OL gain according to the steady-state error coefficient.

(2) Calculate the phase margin of the gain-compensated system with K included.

(3) Determine the required phase lead angle j , and take $j_m = j + (5 \text{ to } 10)°$.

- If w_c is specified, calculate the required phase lead angle at w_c for the gain-compensated system;
- if w_c is not specified, calculate only the difference between the required phase margin and the current phase angle and take it as the required phase lead angle.
- An additional K_c (usually $K_c \geq 1/a$) is adopted when a lead compensator is added to a plant, which causes the Log-magnitude frequency curve to move up. In the end, the w_c after compensation will move to the right to the expected value and actual phase margin will be lower than the the margin we expected.

That is the reason why we take $j_m = j + (5 \text{ to } 10)°$.

(4) Calculate a

$$a = \frac{1 - \sin j_m}{1 + \sin j_m}$$

(5) Calculate $T = \frac{1}{\sqrt{a}\,w_m}$, $aT = \frac{\sqrt{a}}{w_m}$

♦ More notes about the selection of w_m

◊ If w_c is given, take $w_m = w_c$;

◊ If w_c is not given, the selection of w_c is not unique. There could exist different w_m for the same g :

-- When $\arg\left[G_p(\mathrm{j}w)\right]$ is flat, a w_m higher than the expected value can be selected;

-- Set w_m as the future w_{gc}, then the maximum phase angle provided by the compensator will be used for phase compensation.

-- In the case of $w_{gc} = w_m$,
lower w_m will usually give higher g and higher w_m will usually give lower g .

(6) Adjust gain to keep K_v **unchanged.**

(7) Compensated system check and

analysis.

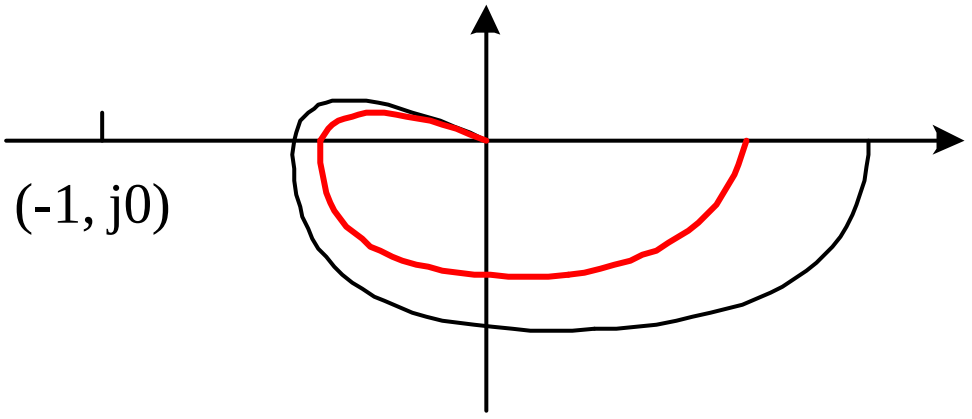
2.5 Phase Lead Compensation by Using the Nyquist Diagram

$$G_c(s) = K_c a \frac{1+Ts}{1+aTs} = K_c \frac{s+\frac{1}{T}}{s+\frac{1}{aT}}, \quad a<1$$

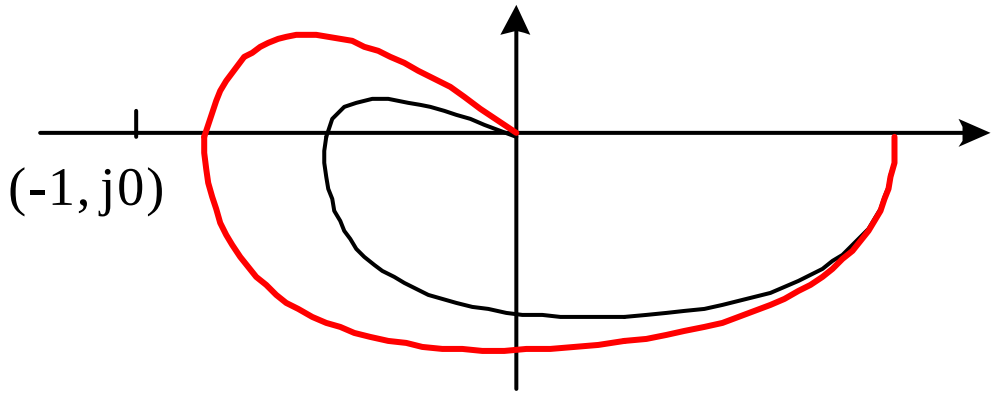
1. Pictorial Explanation

(1) Type 0 systems

¨ $K_c = 1$, K_p –

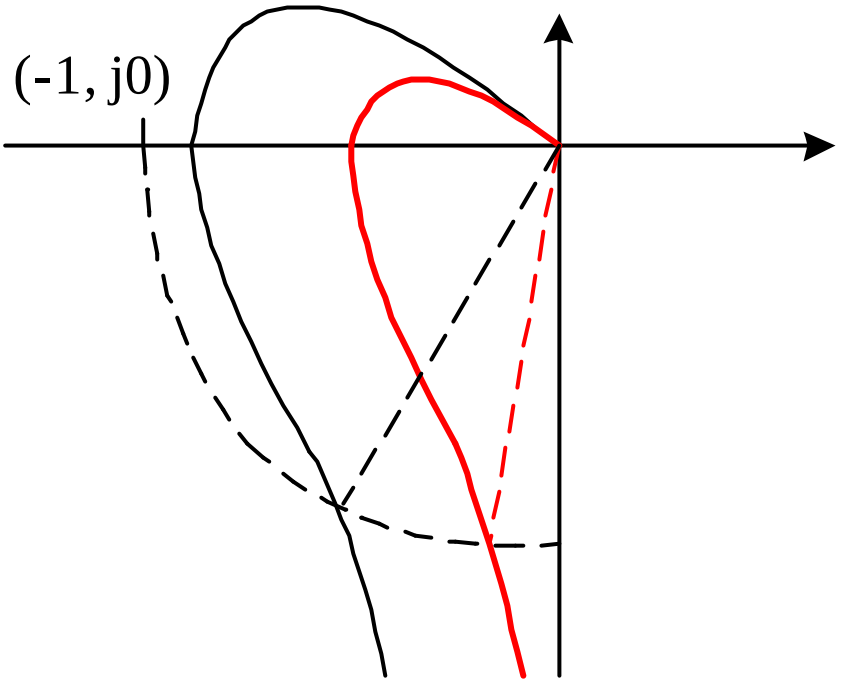


¨ $K_c = \frac{1}{a} > 1$, K_g –, g –



¨ **Phase lead compensator is seldom used for the plant of type 0, unless the plant possesses very high K_p or very high g .**

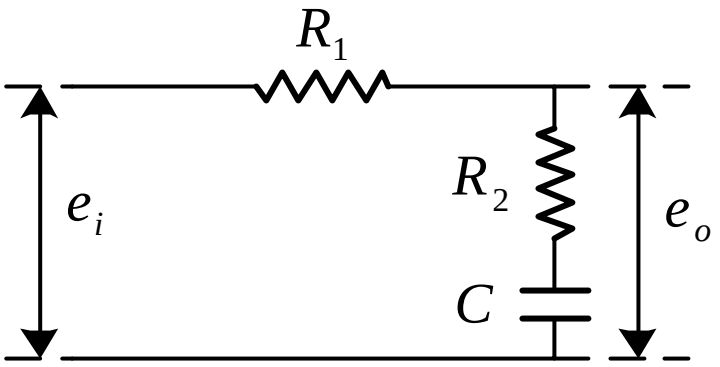
(2) Type 1 systems



¨ g -

§ 3 LAG COMPENSTOR

3.1 Phase Lag Network



$$\frac{E_o(s)}{E_i(s)} = \frac{R_2 C s + 1}{(R_1 + R_2)Cs + 1}$$

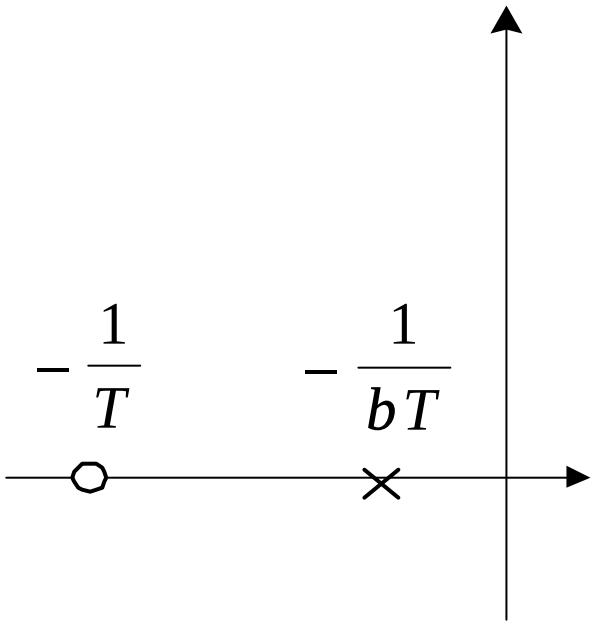
Let $R_2 C = T$, $\frac{R_1 + R_2}{R_2} = b > 1$, **then**

$$\therefore \quad \frac{E_o(s)}{E_i(s)} = \frac{Ts+1}{bTs+1} = \frac{1}{b} \cdot \frac{s + \frac{1}{T}}{s + \frac{1}{bT}}$$

3.2 Characteristics

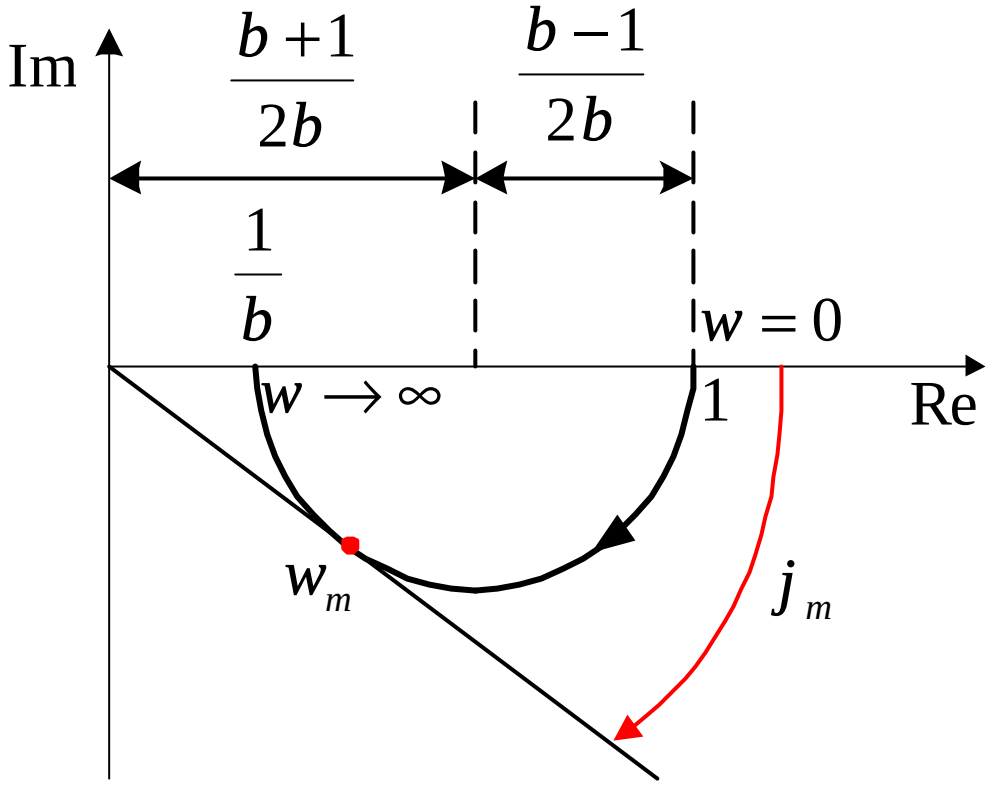
$$G_c(s) = \frac{1+Ts}{1+bTs} = \frac{1}{b} \cdot \frac{s+\frac{1}{T}}{s+\frac{1}{bT}}, \quad b > 1$$

1. Pole-Zero allocation



¨ **Creates lag angle.**

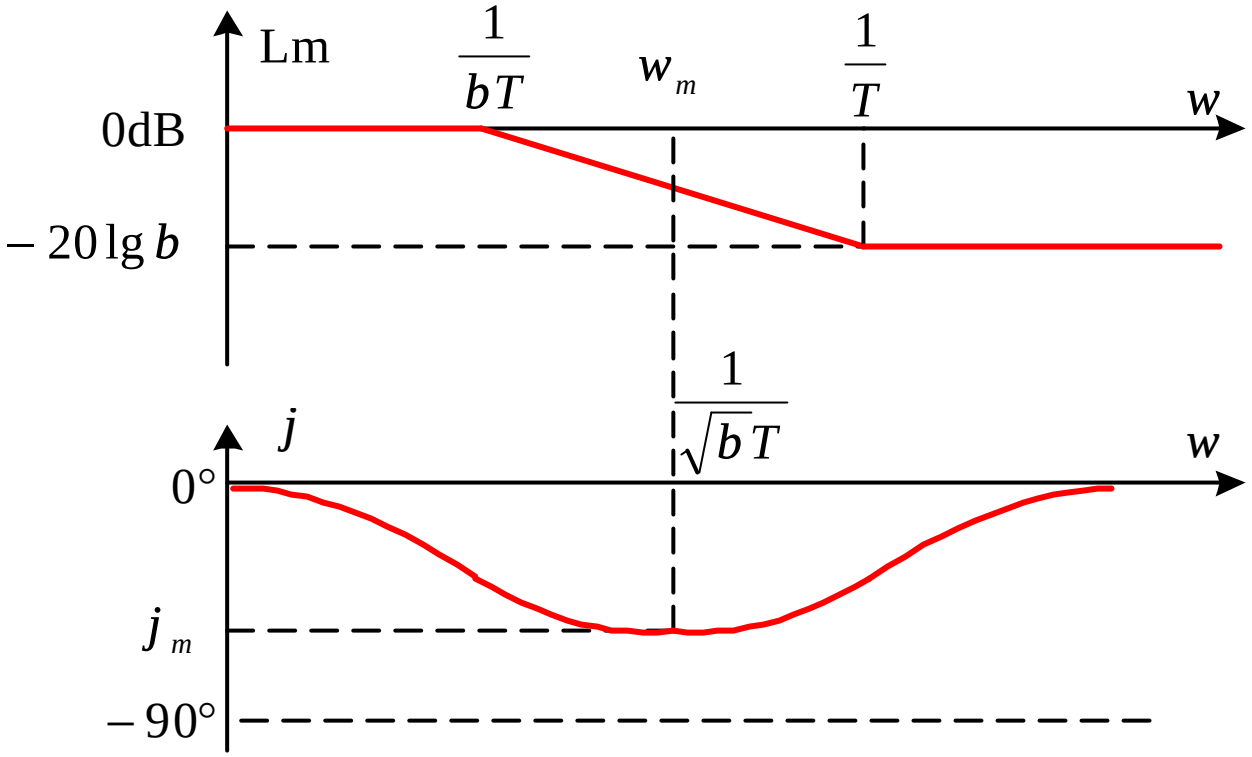
2. Polar plot



$$\sin j_m = \frac{1-b}{1+b}, \quad j_m < 0,$$

$$j_m = \arcsin\frac{1-b}{1+b} = \arctan\frac{1-b}{2\sqrt{b}}$$

3. Bode diagrams



¨ **Behaves like a low-pass filter**

¨ **Additional gain can be used for**

à **Raising the magnitude at low freq. and**

à **Improving the steady-state error**

¨ $w_m = \frac{1}{\sqrt{bT}}$

¨ **Worsens the transient response**

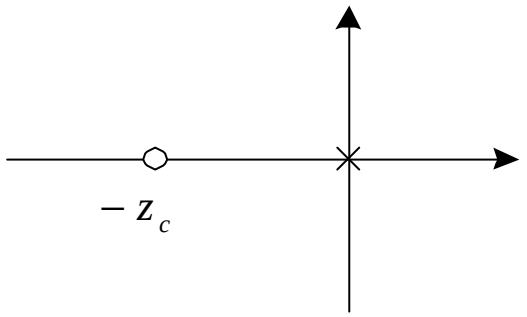
à **Usually added on the low freq. part for**

à **Reducing its influence on the middle freq resp**

3.3 Phase Lag Compensation by Using Root Locus Method

1. Integral Compensator

¨ **A PI controller can improve** e_{ss}
(if original plant has fast trans. resp.)



$$G_c(s) = \frac{s + z_c}{s}$$

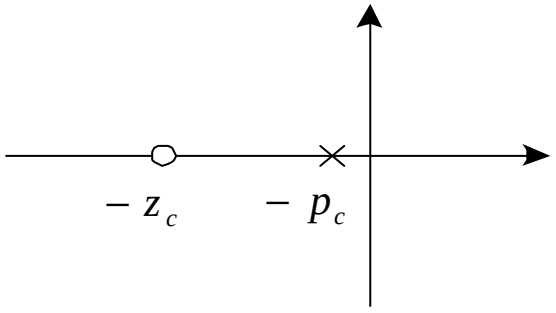
¨ **Feature:**

à **Reduces the steady-state error**

à **Moves the root locus to the right and slows down the response speed**

¨ **Shortcoming:**
Difficult to realize a pure integrator

¨ **Alternative:**
Phase lag compensator with $p_c \ll 1$



$$G_c(s) = \frac{s + z_c}{s + p_c}$$

2. Improvements in Steady-State Error

- **Type 1 system (as example)**

$$G_p(s) = K\frac{\prod_{i=1}^{m}(s+z_i)}{s\prod_{j=1}^{n}(s+p_j)}$$

$$K_v = K\frac{\prod_{i=1}^{m}|z_i|}{\prod_{j=1}^{n}|p_j|}$$

Let

$$G_c(s) = K_c\frac{1}{b}\cdot\frac{s+\frac{1}{T}}{s+\frac{1}{bT}}$$

then

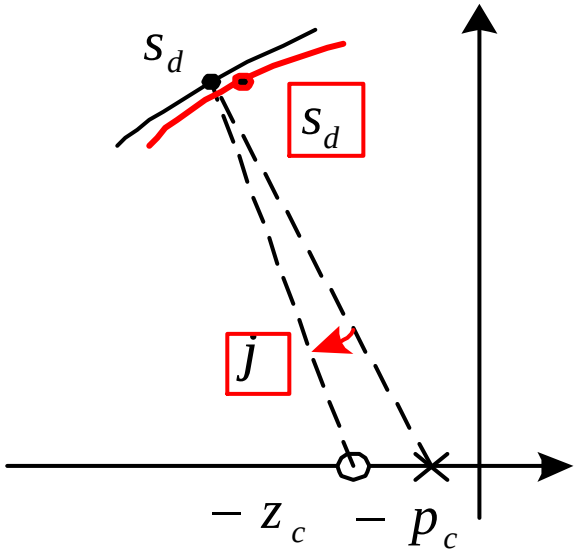
$$G_p(s)G_c(s) = \frac{K\prod_{i=1}^{m}(s+z_i)}{s\prod_{j=1}^{n}(s+p_j)} \cdot \frac{K_c}{b} \cdot \frac{s+\frac{1}{T}}{s+\frac{1}{bT}}$$

$$K_v' = \frac{K\prod_{i=1}^{m}|z_i|}{\prod_{j=1}^{n}|p_j|} \cdot \frac{K_c}{b} \cdot \frac{\frac{1}{T}}{\frac{1}{bT}} = K_v K_c$$

à **Steady-state error coefficient will increase by a factor** K_c

3. Selecting Pole and Zero

(1) p_c/z_c **should be very close to each other Þ Root locus shifts very little**



- $p_c = z_c$, **The root locus will not change**
- $p_c \approx z_c$, **Minor change and dominant poles shift little only**
- $p_c \approx z_c$ **can guarantee that the root locus is almost unchanged in the areas around the dominant poles**

(2) $K_c = b$ Þ **dominant pole unchanged**

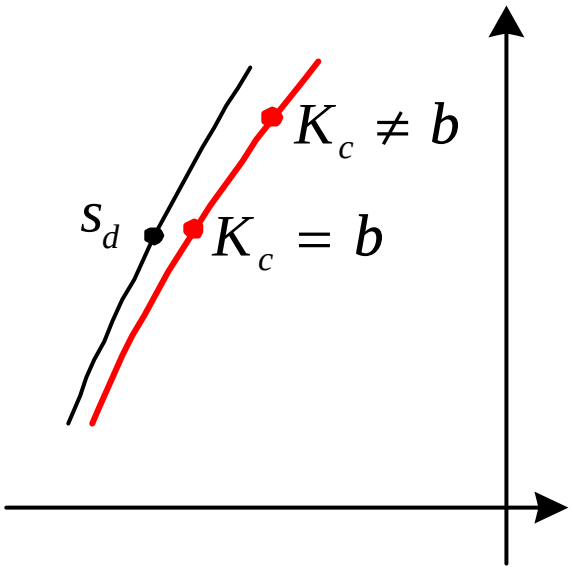
- $p_c \approx z_c$ Þ $\left|\dfrac{s_d + z_c}{s_d + p_c}\right| \approx 1$
- **Magnitude condition** $\left|G_p(s)G_c(s)\right| = 1$

$$\left|G_p(s)G_c(s)\right| = \left|\frac{K\prod_{i=1}^{m}(s_d + z_i)}{s_d\prod_{j=1}^{n}(s_d + p_j)} \cdot \frac{K_c}{b} \cdot \frac{s_d + z_c}{s_d + p_c}\right|$$

$$= \left|\frac{K\prod_{i=1}^{m}(s_d + z_i)}{s_d\prod_{j=1}^{n}(s_d + p_j)}\right| \cdot \frac{K_c}{b} \cdot \left|\frac{s_d + z_c}{s_d + p_c}\right|$$

$$\approx 1 \cdot \frac{K_c}{b} \cdot 1 = 1$$

- $K_c = b$



- $K_c = b$, s_d **remains almost at the original place.**
- $K_c \neq b$, s_d **will move to a new place along the new root locus.**

(3) p_c / z_c **should be very close to the origin**

$z_c \approx p_c$ **and** $z_c / p_c = b >> 1$

Þ $z_c << 1$ **and** $p_c << 1$

4. Example

Example 3.1 Given

$$G_p(s) = \frac{1.06}{s(s+1)(s+2)}$$

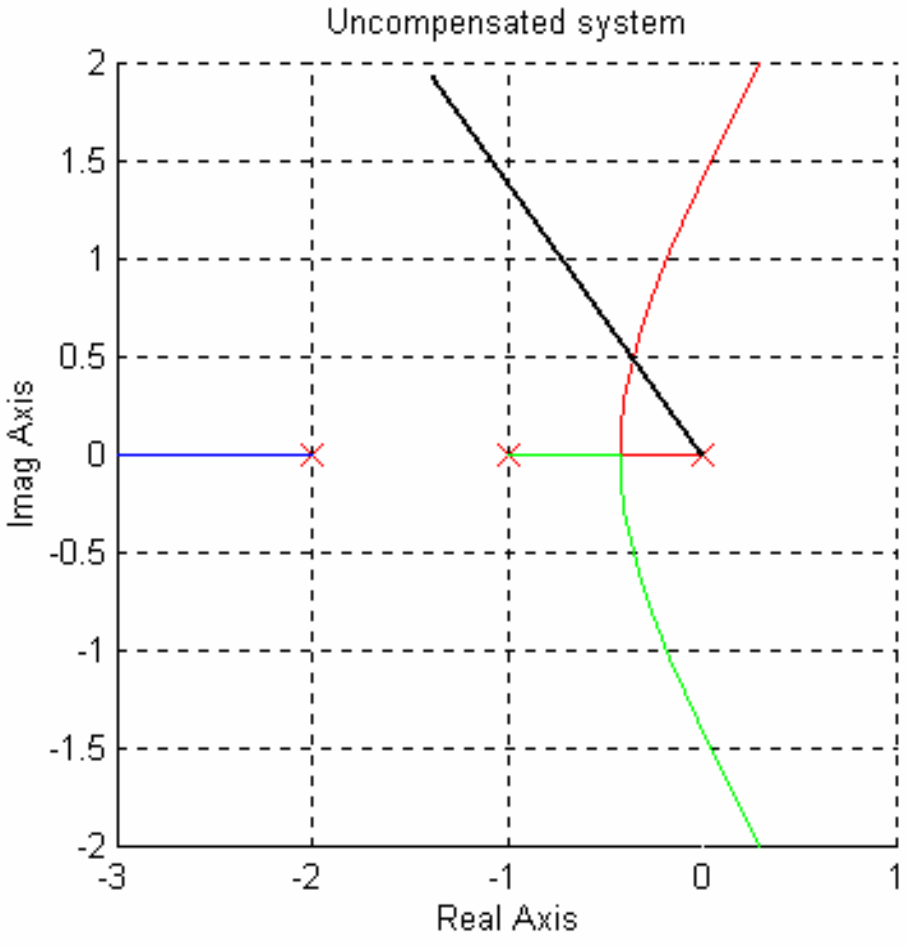
Design a phase lag compensator

Compensated system:

$K_v = 5\ \mathrm{s}^{-1}$, s_d **unchanged**

Solution:

(i) Uncompensated system analysis



$$G_{cl}(s) = \frac{1.06}{s^3 + 3s^2 + 2s + 1.06}$$

$s_{1,2} = -0.33 \pm j0.5$ **(**$w_n = 0.67$ rad/s**,** $z = 0.5$**),**

$s_3 = -2.3$**,** $K_v = 0.53\ \text{s}^{-1}$

(ii) Required additional gain

$$K' = \frac{5}{0.53} = 9.43 \approx 10$$

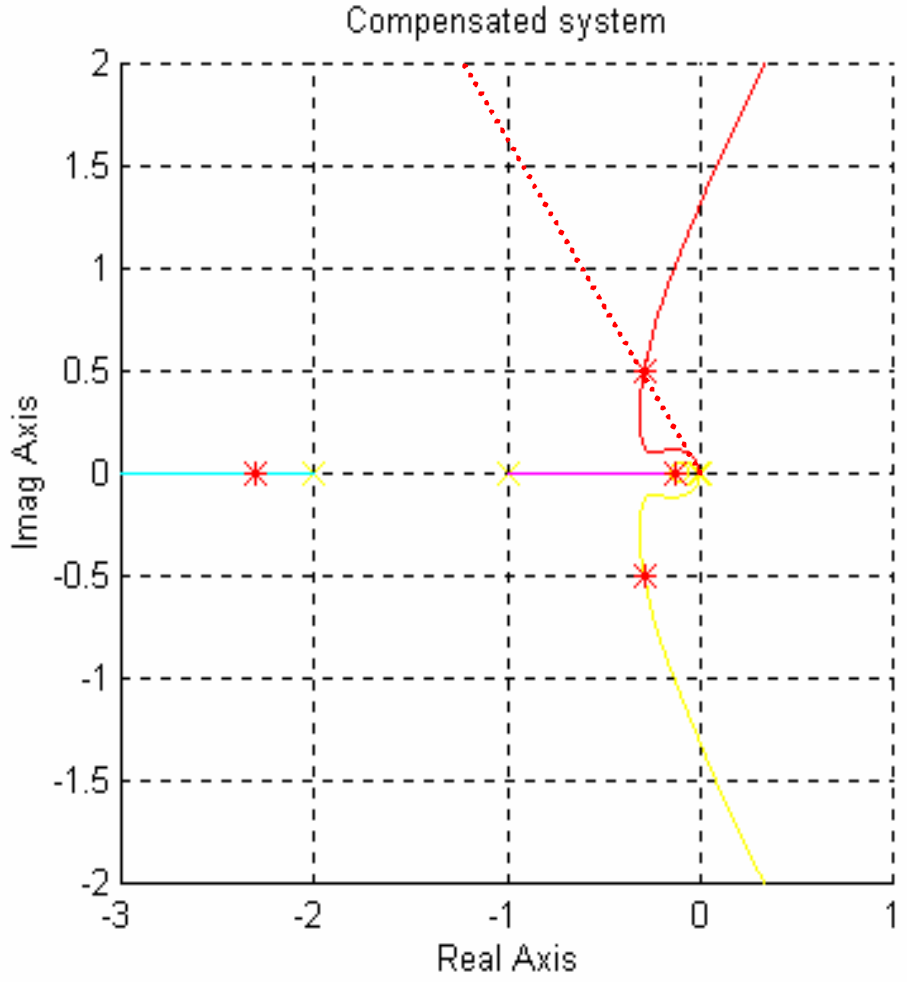
à **A lag compensator is required for raising** K_v **and attaining** s_d **unchanged**

(iii) $b = K' = 10$**,** $-z_c = -0.1$**,** $-p_c = -0.01$

$$G_c(s) = \frac{K_c}{10} \cdot \frac{s + 0.1}{s + 0.01}$$

$$G_p(s)G_c(s) = \frac{1.06K_c}{10} \cdot \frac{s + 0.1}{s(s+1)(s+2)(s+0.01)}$$

à **Root locus of the comp. system**



According to $z = 0.5$ **(by trial and error)**

$$s_{1,2} = -0.2896 \pm j0.5016$$

$$\left|G_p(s_1)G_c(s_1)\right| = 1 \quad \text{Þ}$$

$$\frac{1.06K_c}{10} = \left|\frac{s(s+1)(s+2)(s+0.01)}{s+0.1}\right|_{s=s_1} = 0.9615$$

$$K_c = 9.0708$$

$$G_c(s) = 0.9071\frac{s+0.1}{s+0.01}$$

(iv) Check of the compensated system

$K_v = 4.8075 < 5\ \text{s}^{-1}$,

$s_{1,2} = -0.2896 \pm \text{j}0.5016$ **(dominant poles)**

$s_3 = -0.1243 \approx z_c$

$s_4 = -2.3065 = -7.96 \times 0.2896 = -7.96s$

(v) Further improvements

(i) Smaller $|z_c|$ **with** $b = 10$ **reduces the effect on the root locus**

à **For example**

$$G_{c1}(s) = 0.95\frac{s + 0.05}{s + 0.005} \quad \text{Þ}$$

$$s_{1,2} = -0.314 \pm \text{j}0.544, \quad K_v = 5.04\ \text{s}^{-1}$$

(ii) Higher b **with** z_c **unchanged raises steady-state error coefficient**

à **For example**

$$G_{c2}(s) = 0.905\frac{s + 0.1}{s + 0.008}, \quad b = 12.5 \quad \text{Þ}$$

$$s_{1,2} = -0.2885 \pm \text{j}0.4997, \quad K_v = 5.998\ \text{s}^{-1}$$

”

¨ **Remarks:**

(1) K_v **increases by a factor approximately equal to** b**;**

(2) Small change of s_d **is caused by the distance between** p_c **and** z_c

$$\arg G_c(s_d) = \arg \frac{s+0.1}{s+0.01}\bigg|_{s=-0.33+\mathrm{j}0.59}$$

$$= \arctan \frac{0.59}{-0.23} - \arctan \frac{0.59}{-0.32} = 7.17^\circ$$

$\arg G_{c1}(s_d) = 3.46^\circ$**(**$s_d$ **changes very little)**

(3) The response slows down: w_n **¯,** t_s **-**

5. Procedures (for reference only)

(1) Draw the root locus of $G_p(s)$ **and find** s_d**.**

(2) Calculate the required additional gain.

(3) Determine p_c**,** z_c **and** K_c**.**

(4) Check the dominant poles with the root locus of $G_p(s)G_c(s)$**.**

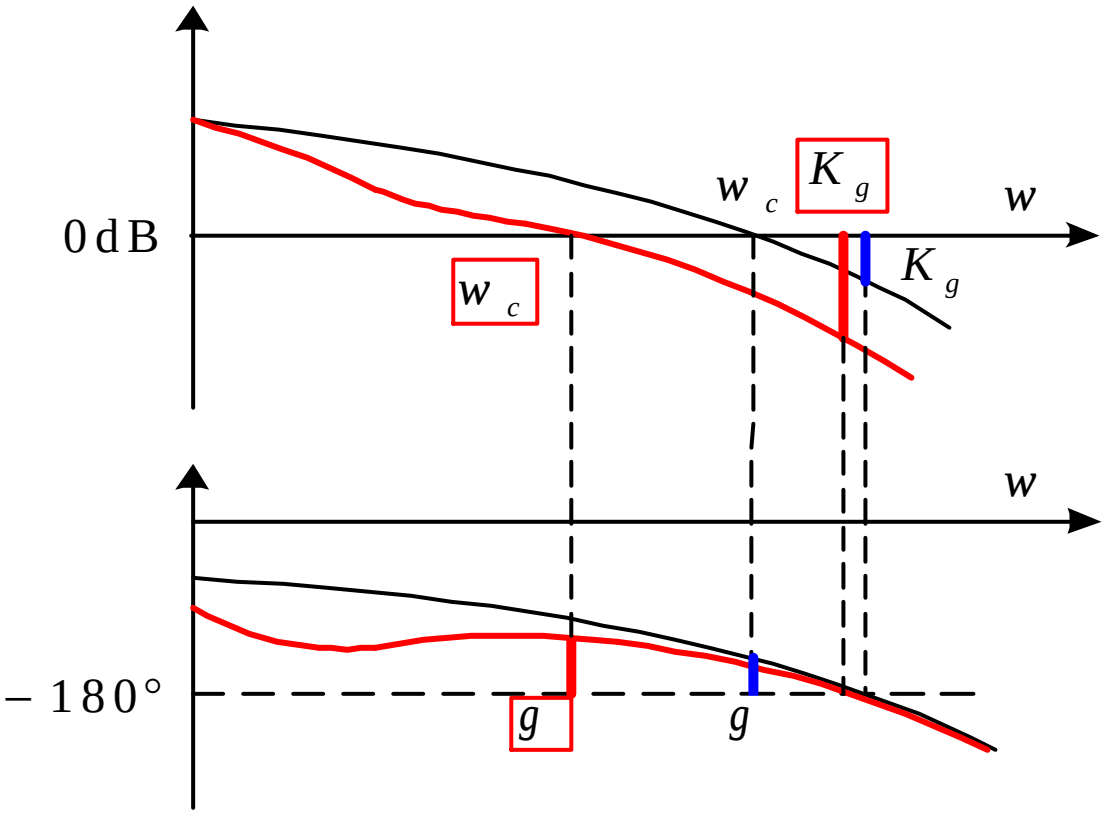
3.4 Phase Lag Compensation by Using Bode Diagrams

1. Selection of $\frac{1}{T}$

$$\frac{1}{T} < \frac{w_c}{10}\left(\frac{w_c}{5}\right) \quad \text{Þ} \quad \left|\arg G_c(jw_c)\right| < 5°$$

2. Pictorial Explanation

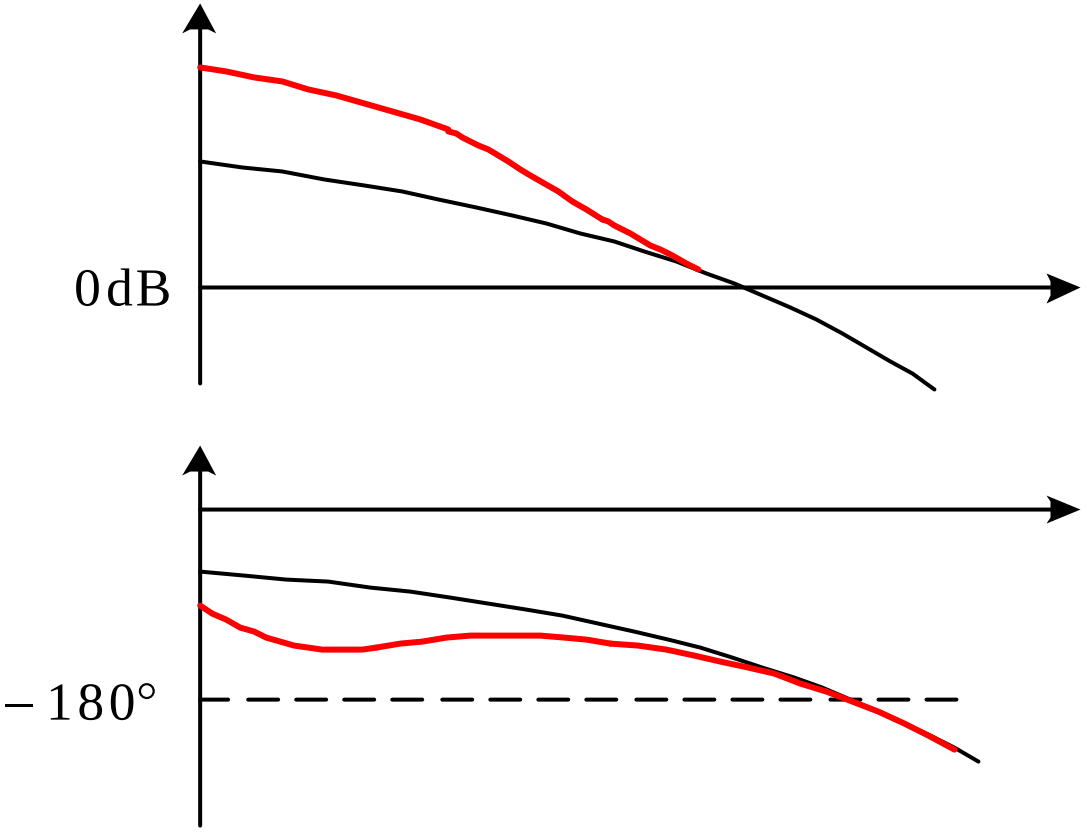
(1) For a plant with high gain



¨ **If there is no additional gain, then**

w_c ¯, g -, K_g -

(ii) For a system with high stability



¨ **If additional gain is allowed, then** K_v -

3. Example

Example 3.2 Given

$$G_p(s) = \frac{K}{s(1+s)(1+0.5s)}$$

Design a cascade compensator such that the compensated system has

$$g \geq 40°, \quad K_g \geq 10 \text{ dB}, \quad K_v \geq 5 \text{ s}^{-1}.$$

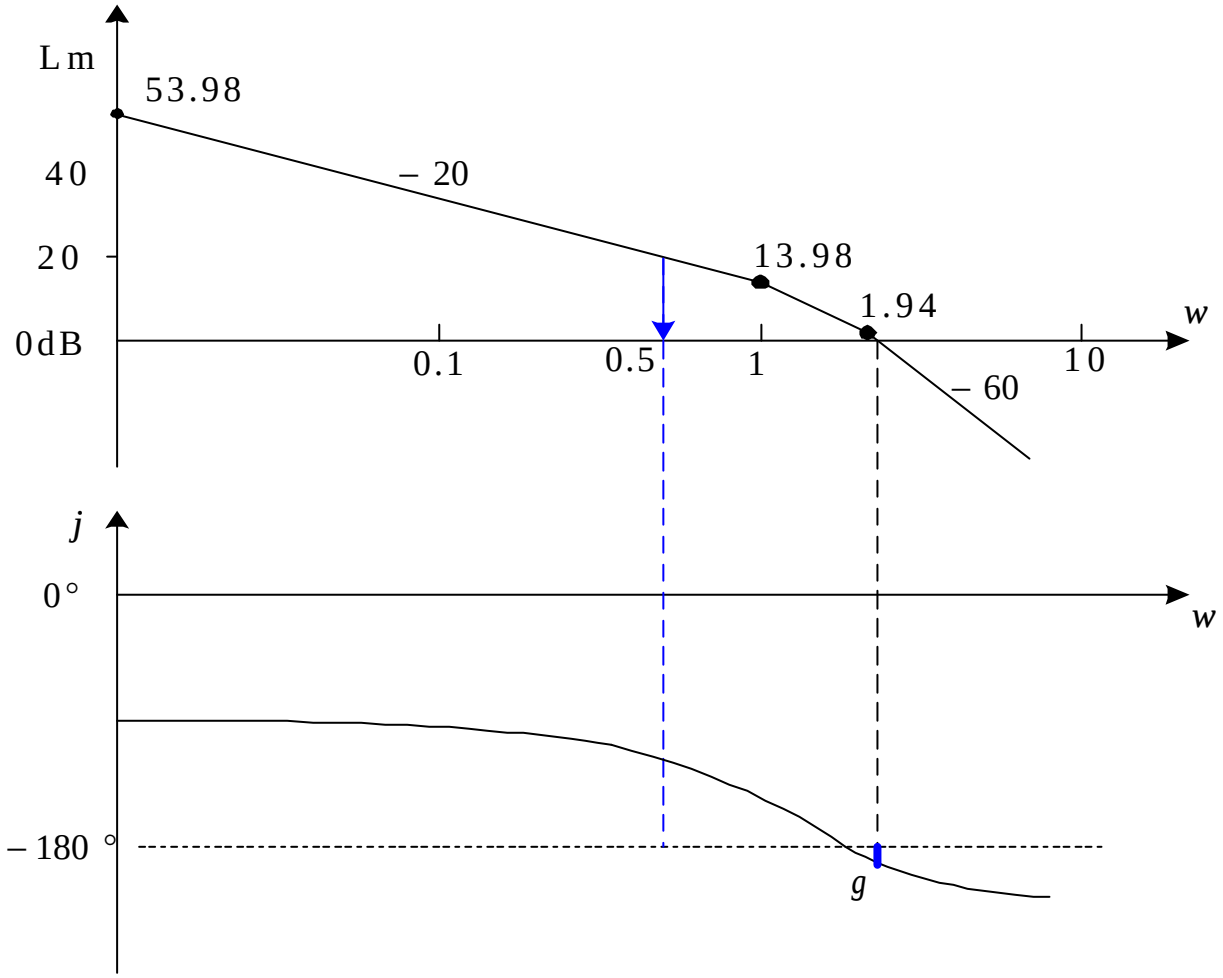
Solution:

(i) Determine *K*

$$K_v = \lim_{s\to 0} sG_p(s) = K = 5$$

$$G_p(s) = \frac{5}{s(1+s)(1+0.5s)}$$

(ii) Gain-compensated system analysis



$w_{gc} = 2.1544$ rad/s, $g = -22.2293°$

$w_{pc} = 1.5$ rad/s, $K_g = -6$ dB

à **The CL system is unstable**

(iii) Selection of w_{gc}

$$g = (40+12)^\circ = 52^\circ$$

$$\arg G_p(\mathrm{j}w) = \arg \frac{1}{(\mathrm{j}w)(1+\mathrm{j}w)(1+\mathrm{j}0.5w)}$$

$$= -90^\circ - \arctan \mathrm{j}w - \arctan \mathrm{j}0.5w = -128^\circ$$

$$w = 0.5 \text{ rad/s}$$

à **Take new crossover freq** $w_{gc} = 0.5$ rad/s

(iv) Let $\frac{1}{T} = \frac{w_{gc}}{5} = 0.1$**,** $T = 10$

(v) Calculate b

$$\left|G_p(\mathrm{j}0.5)\right| \approx 10 = 20 \text{ dB}$$

à **Lag compensator reduces the magnitude at** 0.5 rad/s **by** 20 dB

Þ **To assure** $w_{gc} = 0.5$ rad/s

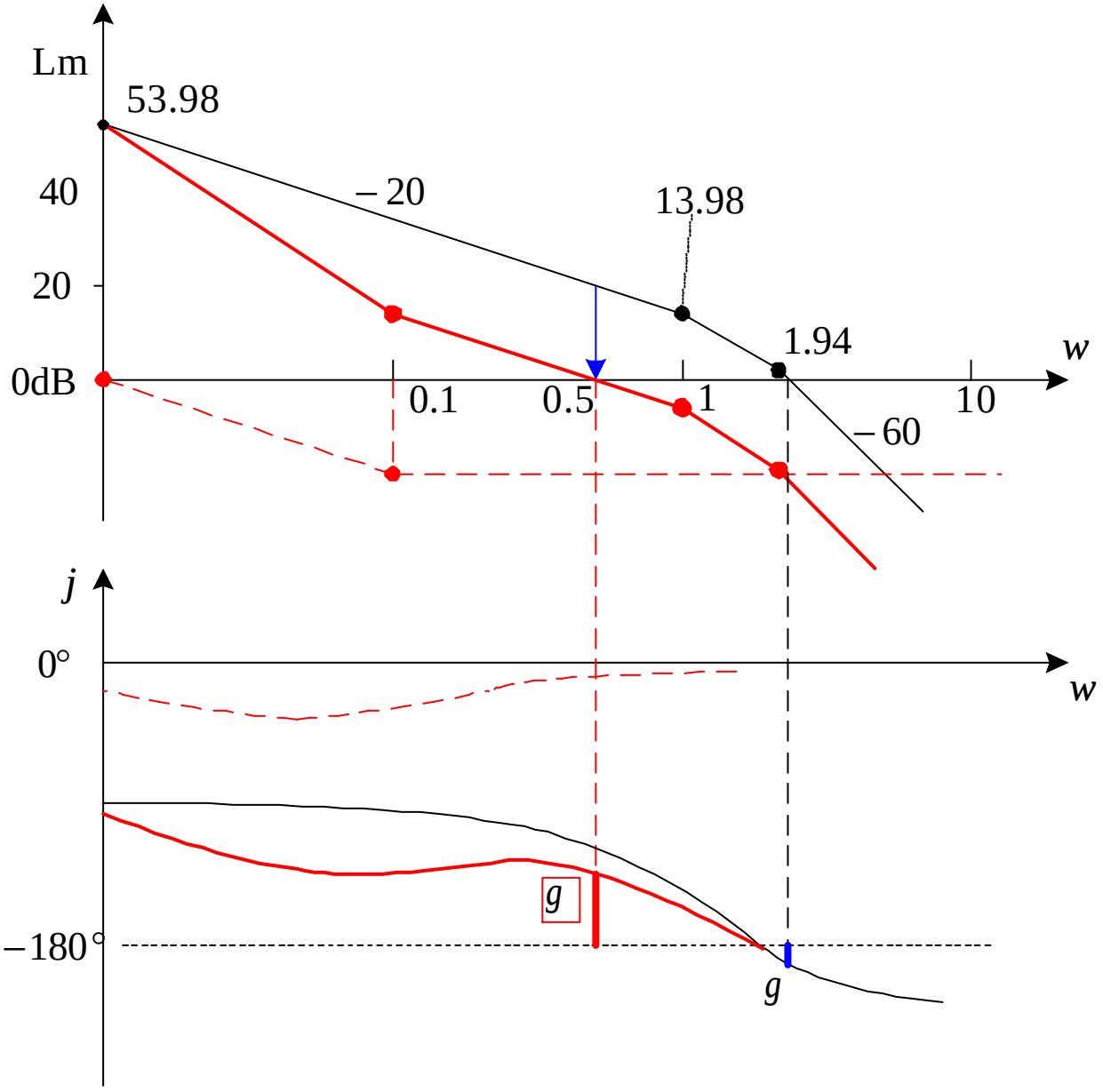
$$-20\lg b = -20, \quad b = 10, \quad \frac{1}{bT} = 0.01$$

$$G_c(s) = \frac{1+Ts}{1+bTs} = \frac{1+10s}{1+100s}$$

or $$G_c(s) = \frac{1}{b}\frac{s+\frac{1}{T}}{s+\frac{1}{bT}} = \frac{1}{10}\cdot\frac{s+0.1}{s+0.01}$$

(vi) Compensated system check

$$G_p(s)G_c(s) = \frac{5(1+10s)}{s(1+s)(1+0.5s)(1+100s)}$$



$K_v = 5\ \mathrm{s}^{-1}$, $w_{gc} = 0.5$ rad/s, $g = 39°$

$w_{pc} = 1.32$ rad/s, $K_g = 11$ dB

(By accurate response.:

$w_{gc} = 0.454$ rad/s, $g = 41.6°$,

$w_{pc} = 1.32$ rad/s, $K_g = 14.3$ dB**.)**

¨ **Remarks:**

(1) Selection of w_{gc}

$$\arg G_p(\mathrm{j}w) + 180° = g + (5 \text{ to } 12)°$$

(2) $\dfrac{1}{T} < \left(\dfrac{1}{5} \text{ to } \dfrac{1}{10}\right) w_{gc}$

(3) Selection of b **is determined by** w_{gc}**, i.e.**

$$20\lg\left|G_p(\mathrm{j}w_{gc})G_c(\mathrm{j}w_{gc})\right| = 0 \text{ dB}$$

3. Procedures (for reference only)

(1) Determine the desired OL gain according to the steady-state error coefficient.

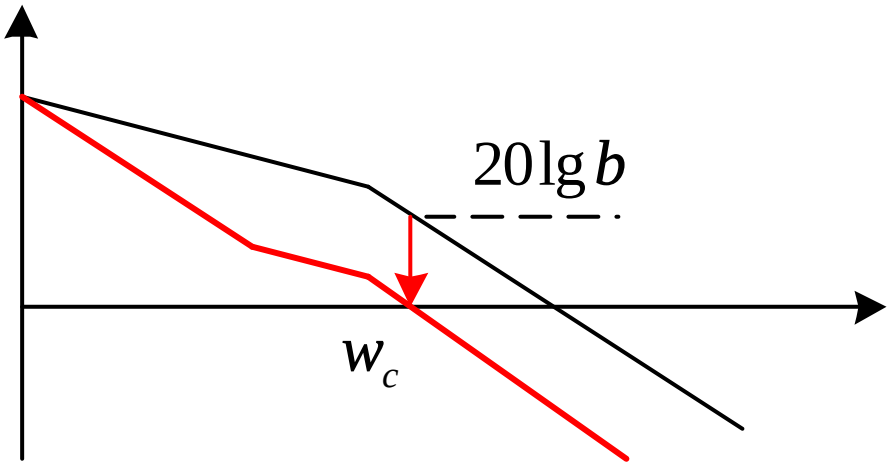
(2) Draw the Bode diagrams of the gain-compensated system and obtain the gain margin and phase margin of the system with *K* included.

(3) Choose a new w_{gc} such that

$$180°+\arg\left[KG_p(jw_{gc})\right]=g+(5\text{ to }12)°.$$

(4) Let $w=\frac{1}{T}<\frac{w_{gc}}{10}$ (at least $w=\frac{1}{T}<\frac{w_{gc}}{2}$)

(5) Calculate b and $\frac{1}{bT}$



The lag compensator should make

$$20\lg\left|G_p(\mathrm{j}w_{gc})G_c(\mathrm{j}w_{gc})\right|=0\ \text{dB}$$

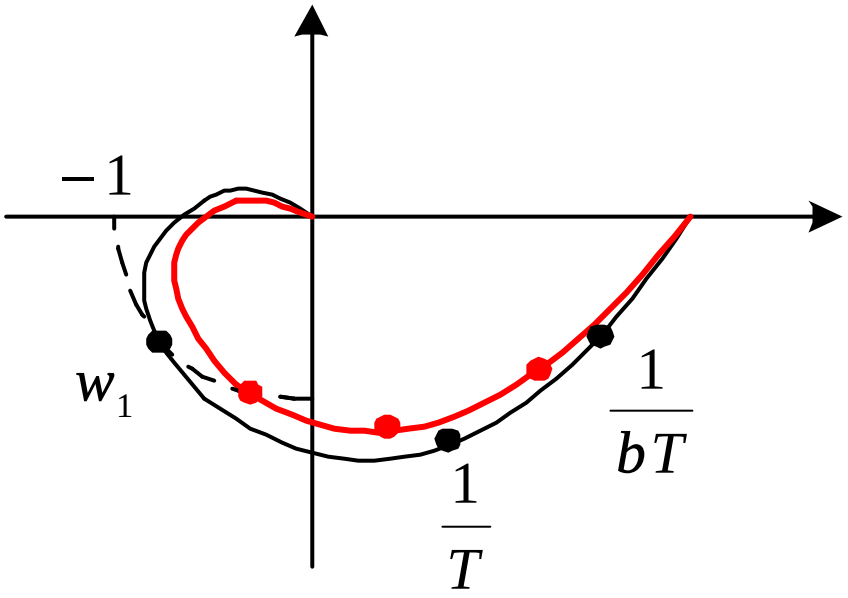
(6) Compensated system check and analysis.

3.5 Phase Lag Compensation by Using the Nyquist Diagram

$$G_c(s) = K_c \frac{1+Ts}{1+bTs}$$

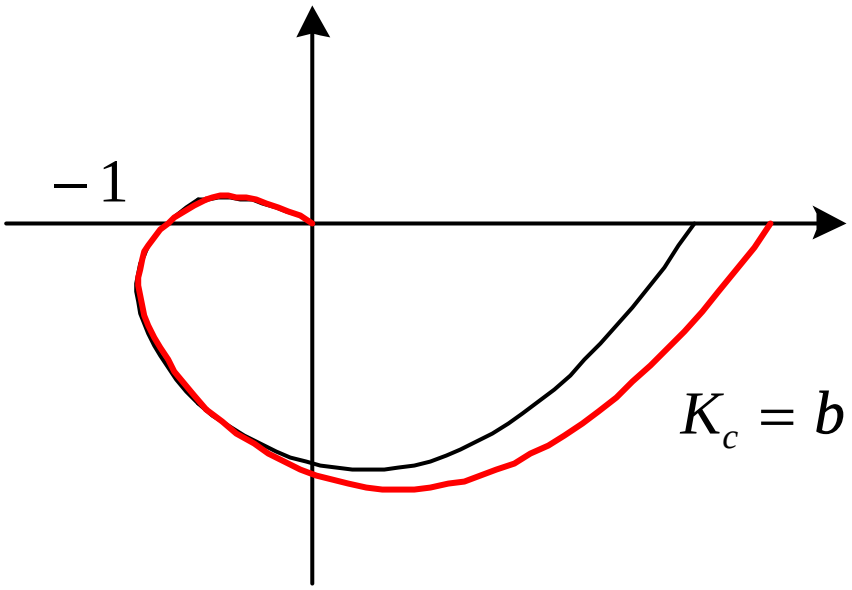
¨ **For type 0 systems (as example)**

1. Improvement in stability (with $K_c = 1$)



¨ K_g -, g - **(Let $\frac{1}{T} \approx \frac{w_1}{10}$)**

2. Reduction of e_{ss}



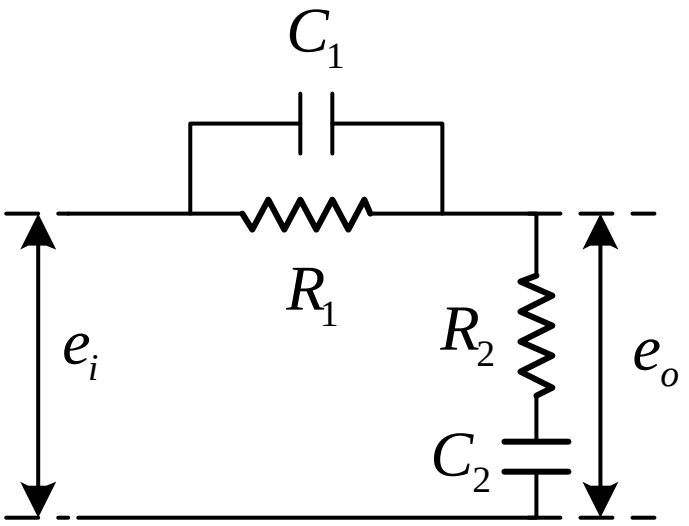
¨ **Additional gain Þ e_{ss} ¯**

§ 4 Lag-Lead Compensator

4.1 Applications

¨ **used for the plants with poor stability and poor steady-state accuracy.**

4.2 Lag-Lead Network



$$\frac{E_o(s)}{E_i(s)} = \frac{(R_1C_1s+1)(R_2C_2s+1)}{(R_1C_1s+1)(R_2C_2s+1)+R_1C_2s}$$

Let $R_1C_1 = T_1$, $R_2C_2 = T_2$,

$$R_1C_1 + R_2C_2 + R_1C_2 = \frac{T_1}{b} + bT_2 \qquad (b > 1)$$

then

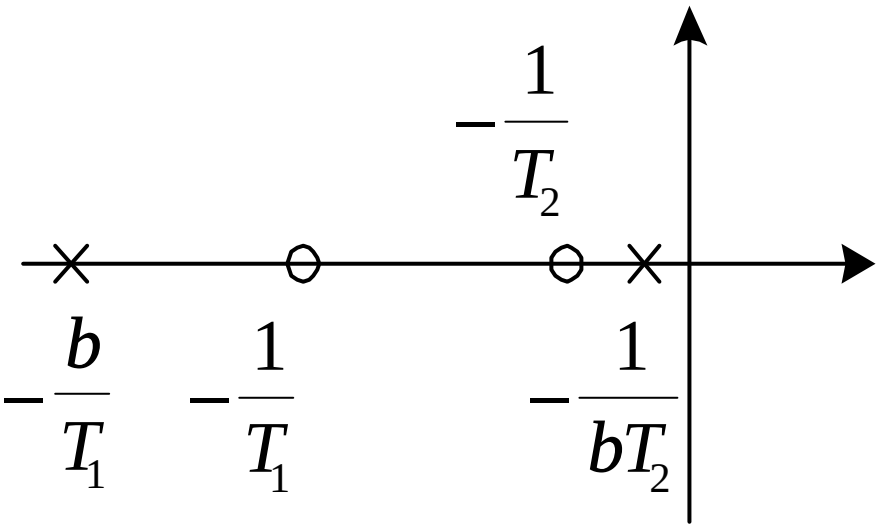
¨

$$\frac{E_o(s)}{E_i(s)} = \frac{(1+T_1 s)(1+T_2 s)}{(1+\frac{T_1}{b}s)(1+bT_2 s)}$$

$$= \frac{\left(s+\frac{1}{T_1}\right)\left(s+\frac{1}{T_2}\right)}{\left(s+\frac{b}{T_1}\right)\left(s+\frac{1}{bT_2}\right)}$$

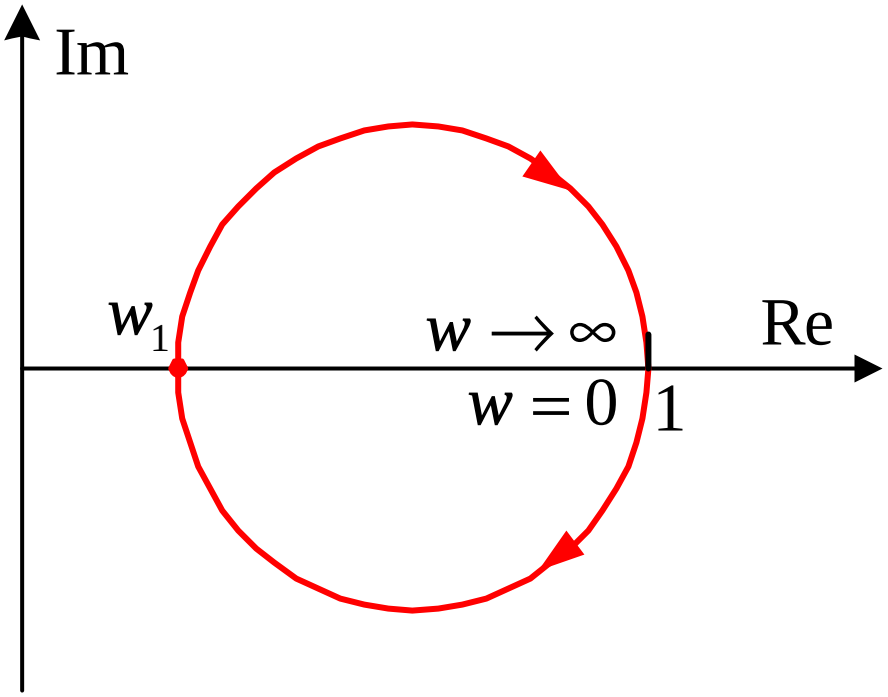
4.3 Characteristics

1. Pole-Zero allocation



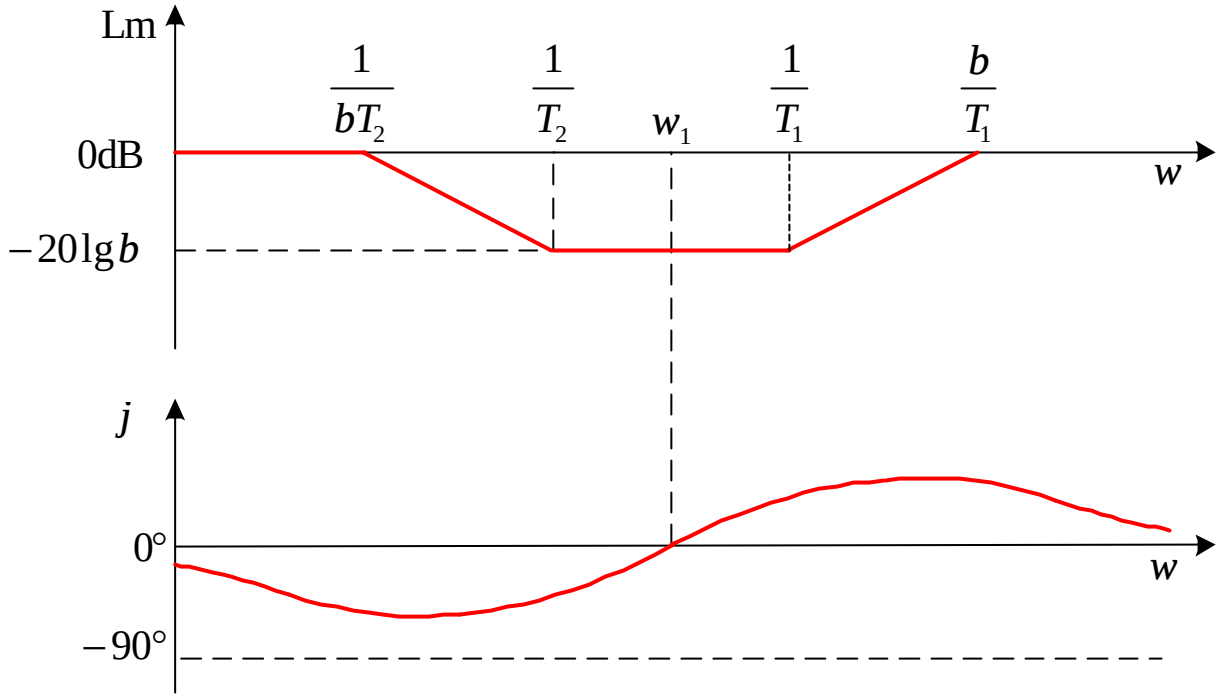
¨ **The two sets of one pole and one zero: similar to [a lead comp. + a lag comp.]**

2. Polar plot



- **When** $w = w_1$, $j = 0$

3. Bode diagrams



- $w_1 = \dfrac{1}{\sqrt{T_1 T_2}}$
- **Lag-lead comp. ≈ Lag comp+Lead comp**
- **Alternate form**

$$G_c(s) = K_c \frac{\left(s+\frac{1}{T_1}\right)}{\left(s+\frac{1}{aT_1}\right)} \cdot \frac{\left(s+\frac{1}{T_2}\right)}{\left(s+\frac{1}{bT_2}\right)}$$

$$= \frac{K_c}{ab} \cdot \frac{(1+T_1 s)}{(1+aT_1 s)} \cdot \frac{(1+T_2 s)}{(1+bT_2 s)}$$

with $a < 1, b > 1, ab \neq 1$

4.4 Lag-Lead Compensation by Using the Root Locus Method

1. Basic Principles

(1) T_1 **and** T_2 **may be selected separately with** $T_2/T_1 >> 1$

(2) $\frac{-1}{T_1}$ **and** $\frac{-b}{T_1}$ **(Lead part):**

- **Function: to improve stability and response speed**
- **Selection:**

à **As for lead compensator**

à $-1/T_1$ **may be selected to cancel a plant pole near to the origin for fast response.**

(3) $\frac{-1}{T_2}$ **and** $\frac{-1}{bT_2}$ **(Lag part):**

- **Function: to improve steady-state accuracy**
- **Selection: Be close to the origin**

2. Example

Example 4.1 $G_p(s) = \dfrac{4}{s(s+0.5)}$

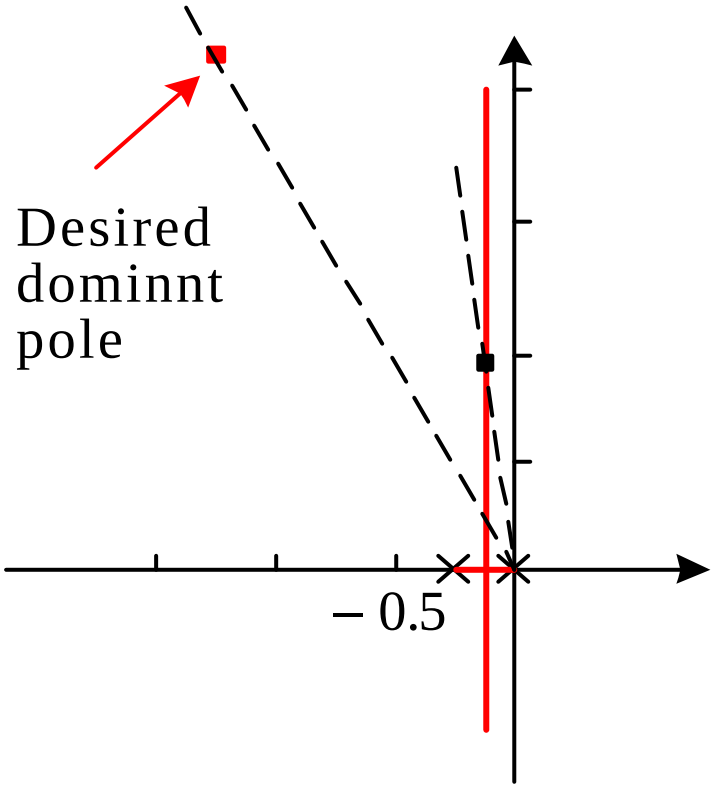
Design a compensator

Compensated system:

$K_v = 50\ \text{s}^{-1},\ z = 0.5,\ w_n = 5$ **rad/s.**

Solution:

(i) Uncompensated system analysis



$K_v = 8\,\text{s}^{-1}$

$z = 0.125$

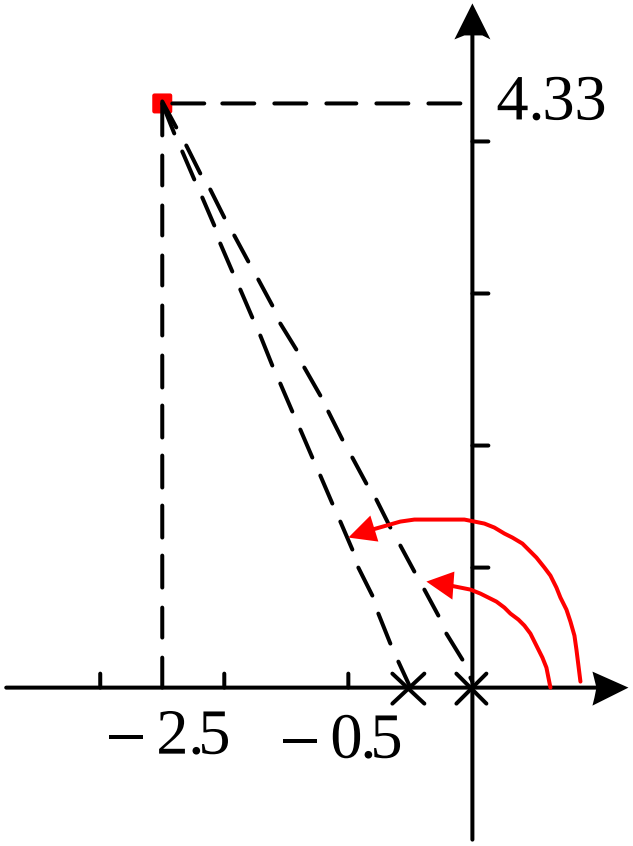
$w_n = 2$ rad/s

$s_{1,2} = -0.25 \pm \text{j}1.984$

(ii) Desired dominant poles

$$s_d = -zw_n \pm \text{j}w_n\sqrt{1-z^2} = -2.5 \pm \text{j}4.33$$

(iii) Required phase lead angle



$$\arg\left[G_p(s_d)\right] = \arg\left[\frac{1}{s(s+0.5)}\right]_{s=s_d}$$

$$= -\arctan\frac{4.33}{-2.5} - \arctan\frac{4.33}{-2.5+0.5}$$

$$= -120° - 114.8° = -234.8°$$

à **Required phase lead angle:**

$$j = -180 - (-234.8°) \approx 55°$$

(iv) Calculate K_c

$$G_c(s) = K_c \frac{\left(s+\frac{1}{T_1}\right)}{\left(s+\frac{b}{T_1}\right)} \cdot \frac{\left(s+\frac{1}{T_2}\right)}{\left(s+\frac{1}{bT_2}\right)}$$

$$K_v = \lim_{s\to 0} sG_p(s)G_c(s) = \lim_{s\to 0} sG_p(s)K_c$$

$$= \lim_{s\to 0} \frac{4K_c}{0.5} = 8K_c = 50$$

$$K_c = 6.25$$

(v) Calculate T_1 and b

à **Assume** $T_2 >> 1$ Þ $\left|\frac{s_d + \frac{1}{T_2}}{s_d + \frac{1}{bT_2}}\right| \approx 1$

à **Magnitude condition:**

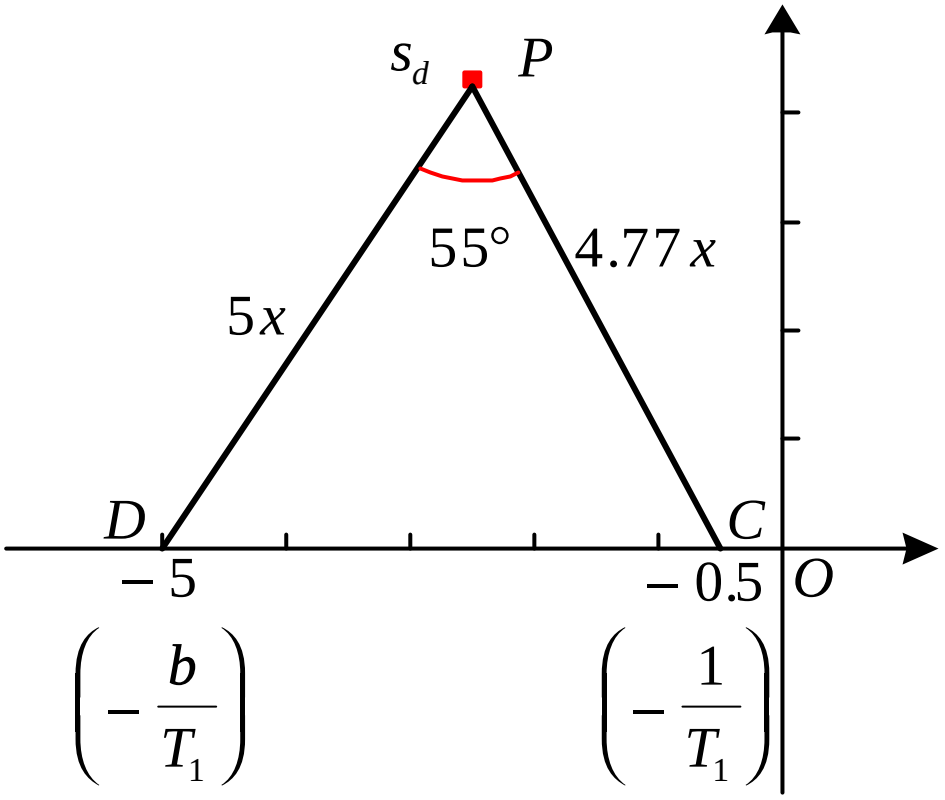
$$\left|G_p(s_d)G_c(s_d)\right| \approx \left|G_p(s_d)\right| \cdot K_c \cdot \left|\frac{s_d + \frac{1}{T_1}}{s_d + \frac{b}{T_1}}\right|$$

$$= \frac{4 \times 6.25}{\left|s_d\right|\left|s_d + 0.5\right|} \cdot \left|\frac{s_d + \frac{1}{T_1}}{s_d + \frac{b}{T_1}}\right|$$

$$= \frac{5}{4.77} \cdot \left|\frac{s_d + \frac{1}{T_1}}{s_d + \frac{b}{T_1}}\right| = 1$$

$$\left|\frac{s_d + \frac{1}{T_1}}{s_d + \frac{b}{T_1}}\right| = \frac{4.77}{5}$$

à **Find** T_1 **by trial and error**



$$\arg\left[\frac{s_d+\frac{1}{T_1}}{s_d+\frac{b}{T_1}}\right]=55° \quad \textbf{i.e} \quad \angle DPC=55°$$

$$\left|\frac{s_d+\frac{1}{T_1}}{s_d+\frac{b}{T_1}}\right|=\frac{4.77}{5} \quad \textbf{i.e.} \quad \frac{|\overline{PC}|}{|\overline{PD}|}=\frac{4.77}{5}$$

Þ $\quad |\overline{CO}|=0.5, \quad |\overline{DO}|=5$

Þ $-\frac{1}{T_1} = -0.5, \; -\frac{b}{T_1} = -5$

i.e. $T_1 = 2, \; b = 10$

(vi) Select T_2 such that

$$\left| \frac{s_d + \frac{1}{T_2}}{s_d + \frac{1}{bT_2}} \right| \approx 1, \;\; 0° < \arg\left[\frac{s_d + \frac{1}{T_2}}{s_d + \frac{1}{bT_2}} \right] < 3°$$

Take $T_2 = 10,$

$$\left| \frac{s_d + 0.1}{s_d + 0.01} \right| \approx 0.9911, \;\; \arg\left[\frac{s_d + 0.1}{s_d + 0.01} \right] = 0.9°$$

Finally

$$G_c(s) = 6.25 \frac{(s+0.5)(s+0.1)}{(s+5)(s+0.01)}$$

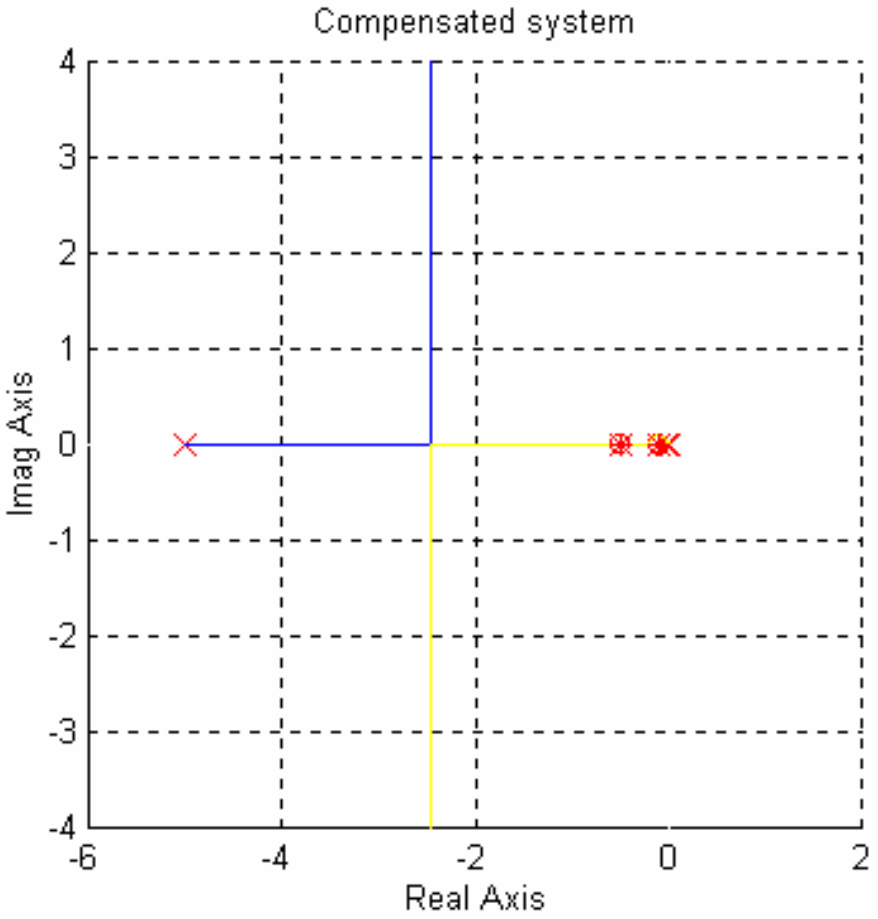
(vii) Check of the compensated system

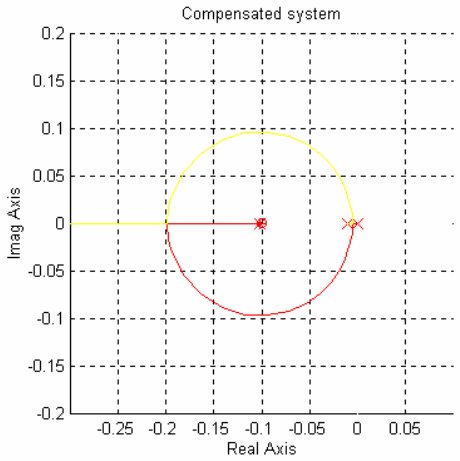
$$G_p(s)G_c(s) = \frac{25(s+0.1)}{s(s+5)(s+0.01)}$$

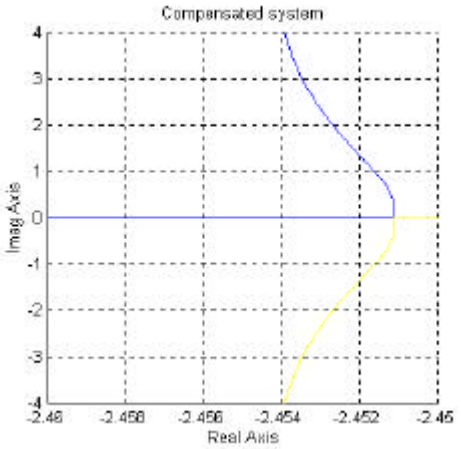
$s_d = -2.454 \pm \text{j}4.304$

$s_3 = -0.1018 \approx -1/T_2, \;\; s_4 = -0.5 = -1/T_1$

$K_v = 50 \text{ s}^{-1}$







Root locus around the origin **Root locus near the break-in point**

Root locus for Example 4.2

¨ **Remarks:**

(1) Cancellation of one plant pole and zero

of lead part is used for fast response.

(2) Separate design of lag and lead parts may be used for this example as well.

3. Procedures (for reference only)

(1) Select the allocation of the desired dominant pole s_d**.**

(2) Calculate the necessary phase angle increment j **to be offered by the lead part.**

(3) Let

$$G_c(s) = K_c \frac{\left(s+\frac{1}{T_1}\right)}{\left(s+\frac{b}{T_1}\right)} \cdot \frac{\left(s+\frac{1}{T_2}\right)}{\left(s+\frac{1}{bT_2}\right)}$$

Decide K_c **to meet the error coefficient.**

(4) Calculate T_1 and b according to

$$\left|G_p(s_d)\right| \cdot K_c \cdot \left|\frac{s_d + \frac{1}{T_1}}{s_d + \frac{b}{T_1}}\right| = 1$$

and

$$\arg\left[\frac{s_d + \frac{1}{T_1}}{s_d + \frac{b}{T_1}}\right] = j$$

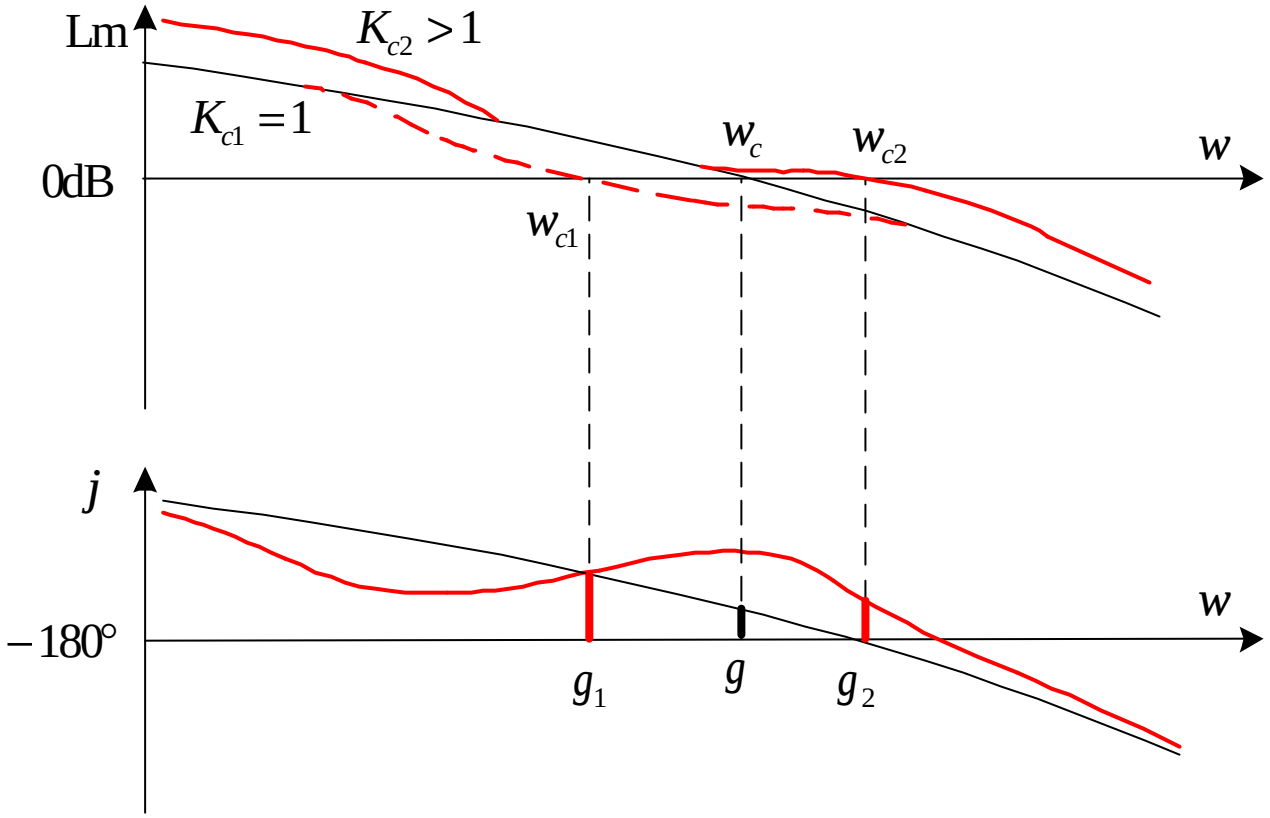
(5) Find T_2 such that

$$\left|\frac{s_d + \frac{1}{T_2}}{s_d + \frac{1}{bT_2}}\right| \approx 1, \quad 0° < \arg\left[\frac{s_d + \frac{1}{T_2}}{s_d + \frac{1}{bT_2}}\right] < 3°$$

(6) Compensated system check.

4.5 Lag-Lead Compensation by Using the Bode Diagrams

1. Pictorial Explanation



- $K_{c1}=1$, w_{c1} ¯, g_1 -, K_g -, K_v **unchanged**
- $K_{c2}>1$, w_{c2} -, g_2 -, K_v -

2. Design methods

(1) Method A: Separated design of the lag and lead parts

- **Select** $\frac{1}{T_2} < \frac{w_c}{10}$ **such that**

$$\arg\left[\frac{\mathrm{j}w_c + \frac{1}{T_2}}{\mathrm{j}w_c + \frac{1}{bT_2}}\right] = -(3^\circ \sim 5^\circ)$$

¨ **Select T_1 such that $1/T_1$ cancels one of the plant poles or creates higher w_{gc}**

(2) Method B: Choose T_1, T_2 and b by experience

¨ **For example, $b = 10$, $T_2 \geq T_1$, and T_1 is selected as for lead compensators**

3. Example

Example 4.2 Given

$$G_p(s) = \frac{K}{s(1+0.1s)(1+0.01s)}$$

Design a cascade compensator such that the compensated system has

$g \geq 40^\circ$, $w_c = 20$ **rad/s,** $K_v \geq 100\ \mathrm{s}^{-1}$

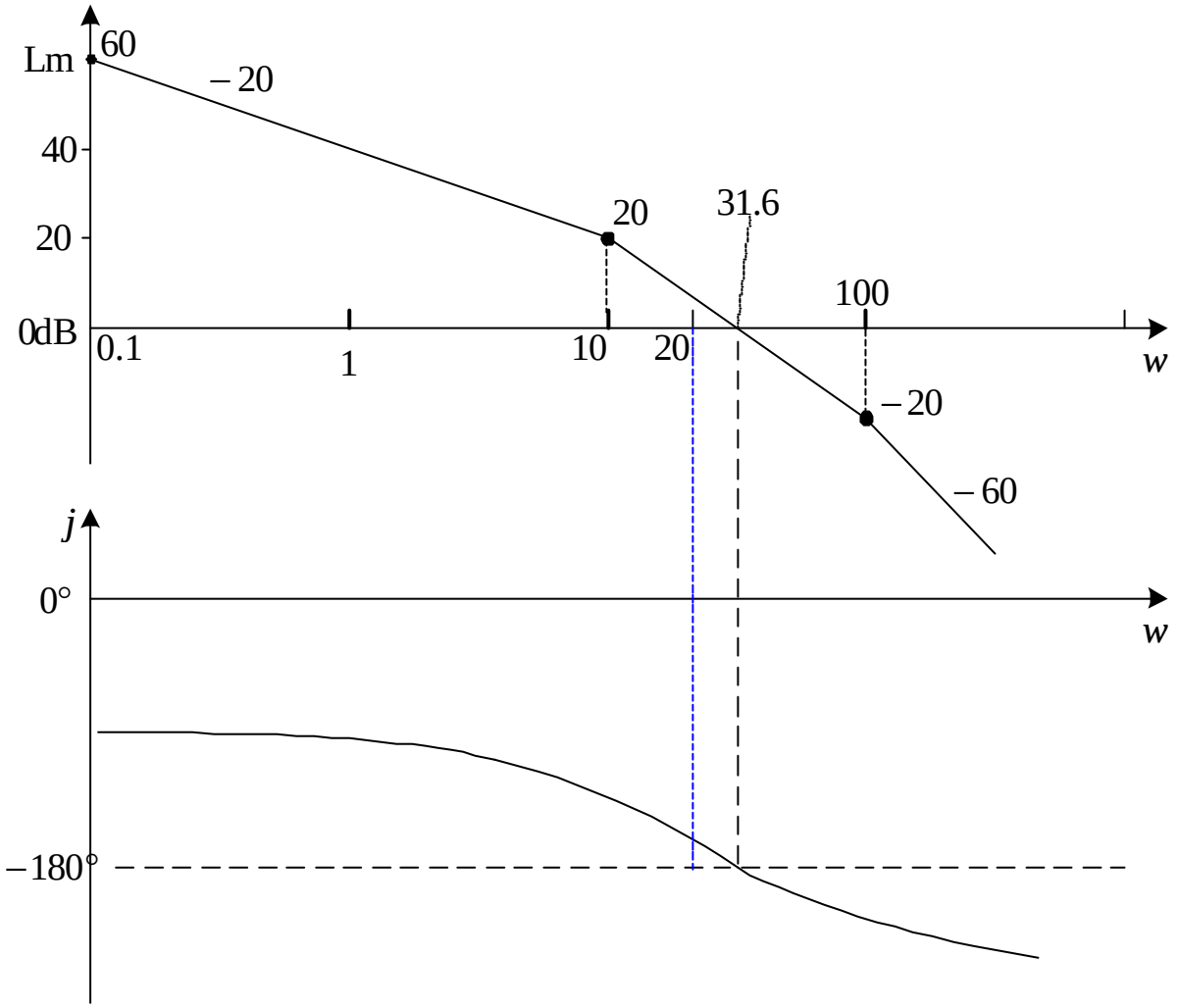
Solution:

(i) Determine the desired *K*.

$$K_v = \lim_{s\to 0} sG_p(s) = K = 100$$

$$G_p(s) = \frac{100}{s(1+0.1s)(1+0.01s)}$$

(ii) Gain-compensated system analysis



$$w_{gc} = 31.6 \text{ rad/s}, \; g = 0°$$

$$w_{pc} = 31.6 \text{ rad/s}, \; K_g = 0.83 \text{ dB}$$

à **The CL system is almost unstable**

à $\arg G(\text{j}20) = -165°$**, Required** $j = 25°$

à **Take** $j_m = 40°$

(iii) Design of the lead part $G_{c1}(s)$

$$a = \frac{1-\sin 40^\circ}{1+\sin 40^\circ} = 0.2174$$

$$\frac{1}{T} = \sqrt{a}w_c = \sqrt{0.2174} \times 20 = 9.33$$

$$T = 0.107, \quad aT = 0.023$$

à **Lead part of the compensator should be**

$$G_{c1}(s) = \frac{1+0.107s}{1+0.023s}$$

à **The final lead part is**

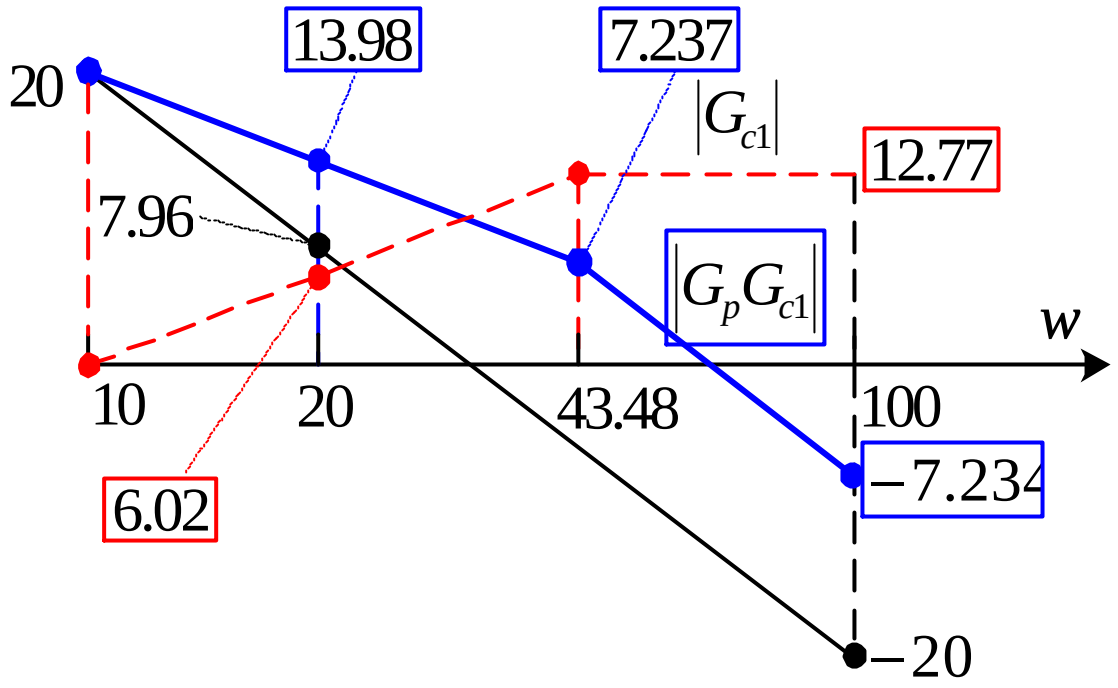
$$G_{c1}(s) = \frac{1+0.1s}{1+0.023s}$$

$$= 4.348\frac{s+10}{s+43.48}, \quad a = 0.23$$

Reason: the zero of the compensator is near to one pole of the plant

(iv) Design of the lag part $G_{c2}(s)$

à **Bode diagrams of** $G_p(\mathrm{j}w)G_{c1}(\mathrm{j}w)$



$$\left|G_p(\mathrm{j}20)G_{c1}(\mathrm{j}20)\right| = 7.96 + 6.02 \approx 14 \ \ \mathrm{dB}$$

à $\left|G_p(\mathrm{j}20)G_{c1}(\mathrm{j}20)G_{c2}(\mathrm{j}20)\right| = 1$

Þ $\left|G_{c2}(\mathrm{j}20)\right| = -20\log b = -14 \ \ \mathrm{dB}$

$b \approx 5$ **NB:** $ab \neq 1$

à **Let** $1/T_2 = w_c/5 = 4$, $1/bT_2 = 0.8$

$T_2 = 0.25$, $bT_2 = 1.25$

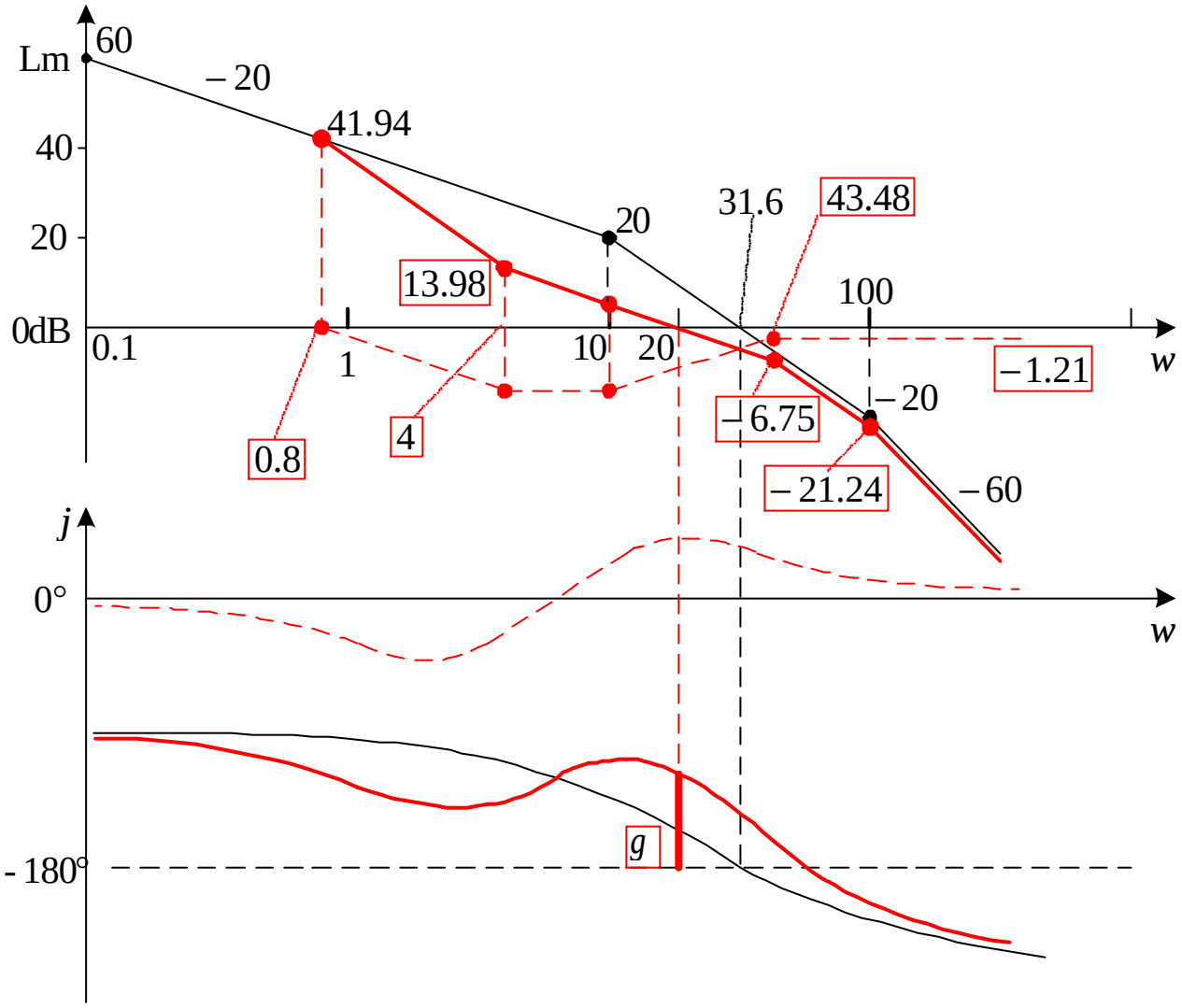
$$G_{c2}(s) = \frac{1+0.25s}{1+1.25s} = \frac{1}{5}\cdot\frac{s+4}{s+0.8}$$

à $$G_c(s) = \frac{(1+0.1s)(1+0.25s)}{(1+0.023s)(1+1.25s)}$$

$$= 0.87\,\frac{(s+10)(s+4)}{(s+43.48)(s+0.8)} \qquad (1/ab = 0.87)$$

(v) Compensated system check

$$G_p(s)G_c(s) = \frac{100(1+0.25s)}{s(1+0.01s)(1+0.023s)(1+1.25s)}$$



$w_{gc} = 20$ rad/s, $g = 44.97°$, $K_v = 100\ \text{s}^{-1}$

$w_{pc} = 62.4$ rad/s, $K_g = 13.05$ dB

(The accurate values:

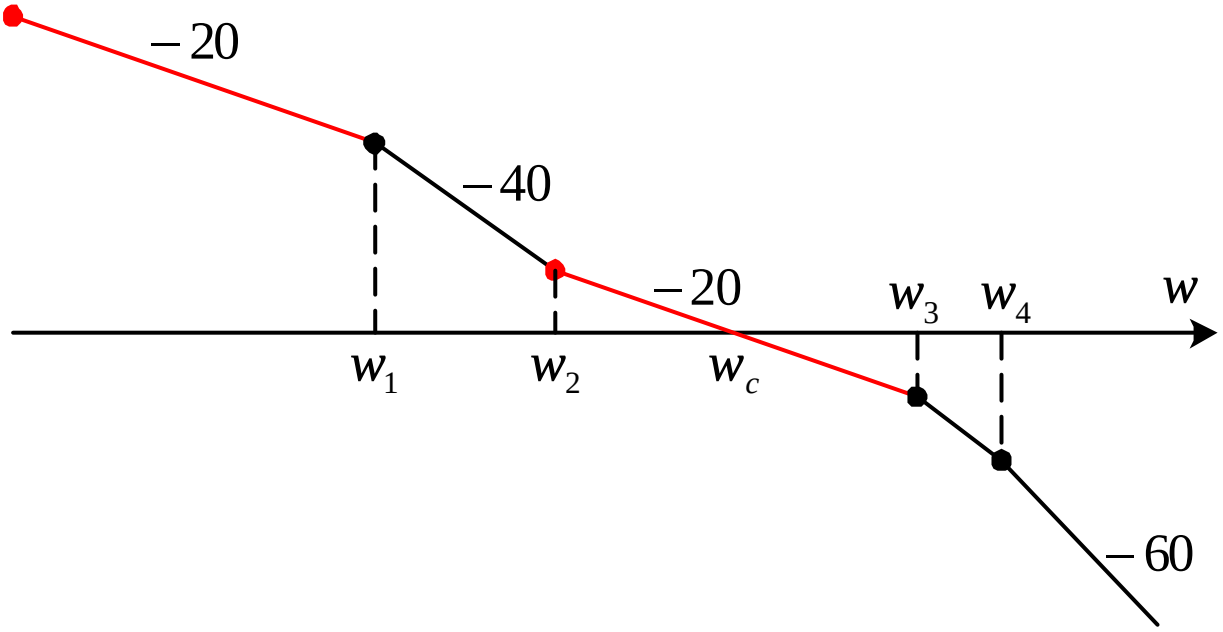
$w_{gc} = 18.4977$ rad/s, $g = 46.7474°$,

$w_{pc} = 62.3624$ rad/s, $K_g = 16.1409$ dB**)**

4.6 Expected Open Loop Frequency Response

¨ **A typical 4th order OL plant**

$$G_p(s)G_c(s) = \frac{K\left(1+\frac{s}{w_2}\right)}{s\left(1+\frac{s}{w_1}\right)\left(1+\frac{s}{w_3}\right)\left(1+\frac{s}{w_4}\right)}$$



(1) Middle frequency range

· **Higher** w_c **Þ Fast response**

· **Slope of -20 dB/decade Þ Stability**

(2) Higher frequency range

· **Lower** w_3 **and** w_4 **Þ Noise reduction**

(3) Lower frequency range

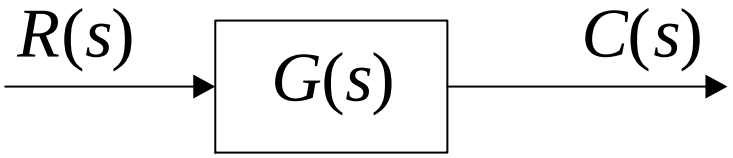
- **Lower** w_2
  - Þ **Better stability**
  - Þ **Response approaching to the steady-state value very slowly**

§ 5 Feedback Compensation

5.1 Advantages of F/B Compensation

1. Reduction of the Influences of Environmental Conditions

¨ **Open loop plant**



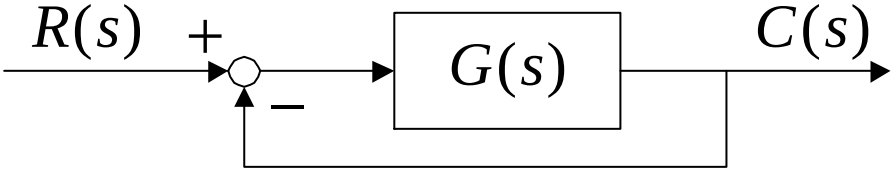
$$C(s) = G(s)R(s)$$

à **Under small change** $\mathrm{d}G(s)$**:**

$$\mathrm{d}C(s) = \mathrm{d}G(s) \cdot R(s) = \frac{\mathrm{d}G(s)}{G(s)} G(s)R(s)$$

$$= \frac{\mathrm{d}G(s)}{G(s)} C(s)$$

¨ **Unity feedback**



$$C(s) = \frac{G(s)}{1 + G(s)} R(s)$$

$$\mathrm{d}C(s) = \frac{[1+G(s)]\mathrm{d}G(s) - G(s)\mathrm{d}G(s)}{[1+G(s)]^2} R(s)$$

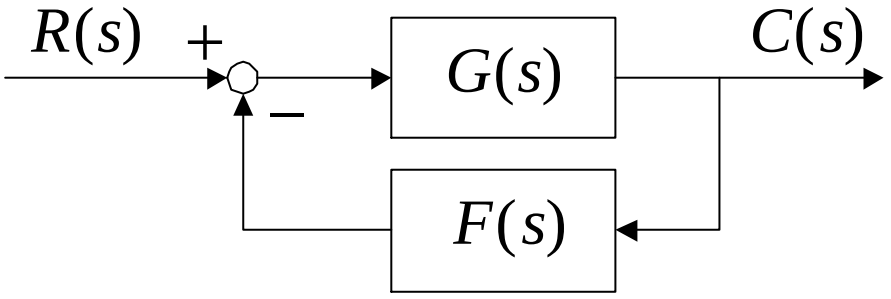
$$= \frac{\mathrm{d}G(s)}{[1+G(s)]^2} R(s)$$

$$= \frac{1}{1+G(s)} \cdot \frac{G(s)}{1+G(s)} \cdot \frac{\mathrm{d}G(s)}{G(s)} R(s)$$

$$= \frac{1}{1+G(s)} \cdot \frac{\mathrm{d}G(s)}{G(s)} C(s)$$

à **Influence is reduced by** $\frac{1}{1+G(s)}$

¨ **Non-unity feedback**



$$C(s) = \frac{G(s)}{1+G(s)F(s)} R(s)$$

à **If** $F(s)$ **does not vary**

$$\mathrm{d}C(s) = \frac{[1+GF]\mathrm{d}G - GF\mathrm{d}G}{[1+GF]^2} R$$

$$= \frac{\mathrm{d}G}{[1+GF]^2} R$$

$$= \frac{1}{1+GF} \cdot \frac{G}{1+GF} \cdot \frac{\mathrm{d}G}{G} R$$

$$= \frac{1}{1+G(s)F(s)} \cdot \frac{\mathrm{d}G(s)}{G(s)} C(s)$$

· $\left|1+G(s)F(s)\right| >> 1$

Þ **Influence of** $\mathrm{d}G(s)$ **will be greatly reduced**

à **If** $F(s)$ **varies, but** $G(s)$ **does not vary**

$$\mathrm{d}C(s) = \frac{[1+GF] \times 0 - G \cdot G\mathrm{d}F}{[1+GF]^2} R$$

$$= \frac{-G^2 \mathrm{d}F}{[1+GF]^2} R$$

$$= -\frac{G\mathrm{d}F}{1+GF} \cdot \frac{G}{1+GF} R$$

$$= -\frac{G(s)\mathrm{d}F(s)}{1+G(s)F(s)} C(s)$$

· $\left|G(s)F(s)\right| >> 1$ Þ $\mathrm{d}C(s) = -\frac{\mathrm{d}F(s)}{F(s)} C(s)$

· $G(s)$ **is designed for stability**

- $F(s)$ **is designed for rejection of the environmental disturbance**

2. Speedup of the Response

Example 5.1 Design different types of compensators for the open loop transfer function

$$G(s) = \frac{K}{1 + 0.1s}$$

such that the steady-state output of the CL system $C(\infty) = 0.9$**.**

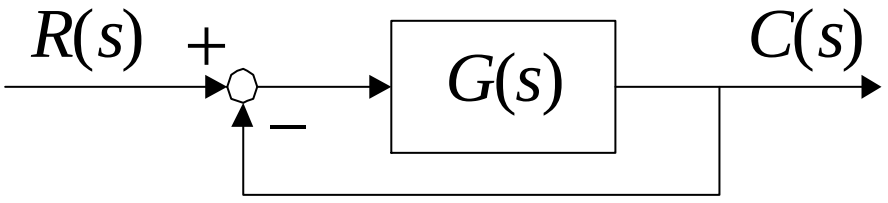
Solution:

(i) OL analysis

$K = 0.9$ Þ $C(\infty) = 0.9$

The system has $T = 0.1$ **sec.**

(ii) Unity feedback



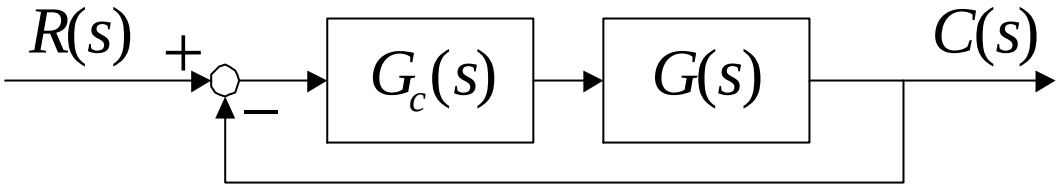
$$C(s) = \frac{K}{1 + K + 0.1s} R(s)$$

$$\frac{C(s)}{R(s)} = \frac{K}{1+K+0.1s}$$

$$K = 9 \quad \text{Þ} \quad C(\infty) = 0.9$$

CL system has $T = 0.01$ **sec.**

(iii) Lead compensation



Suppose $G_c(s) = 0.5\dfrac{1+0.1s}{1+0.05s}$

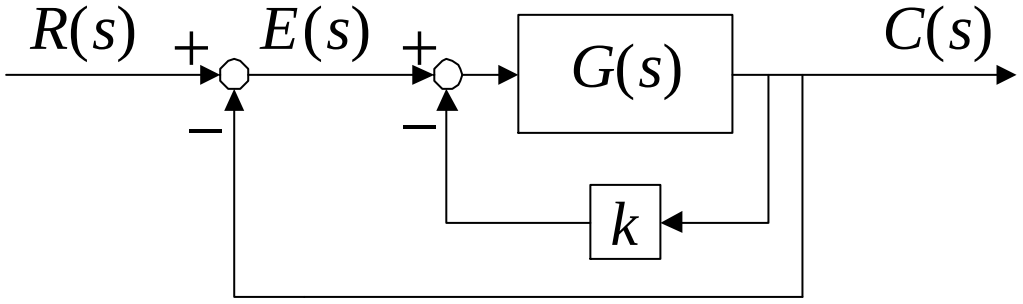
$$G(s)G_c(s) = \frac{0.5K}{1+0.05s}$$

$$\frac{C(s)}{R(s)} = \frac{0.5K}{1+0.5K+0.05s}$$

$$K = 18 \quad \text{Þ} \quad C(\infty) = 0.9$$

CL system has $T = 0.005$ **sec.**

(iv) Feedback compensation



$$\frac{C(s)}{E(s)} = \frac{G(s)}{1+kG(s)} = \frac{K}{1+kK+0.1s}$$

$$\frac{C(s)}{R(s)} = \frac{K}{1+K+kK+0.1s}$$

For $C(\infty) = 0.9$ **and** $T = 0.005$ **sec**

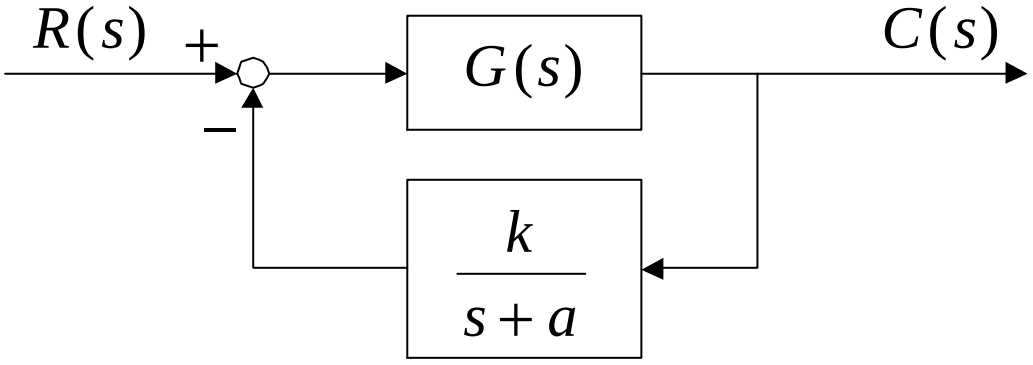
$$\frac{K}{1+K+Kk} = 0.9$$

$$\frac{0.1}{1+K+Kk} = 0.005 = \frac{1}{200}$$

$K = 18$ **and** $k = 0.05556$

à **A simple feedback achieves the same result as a lead compensator.**

(v) Feedback lag compensation



$$\frac{C(s)}{R(s)} = \frac{10K(s+a)}{s^2+(10+a)s+10(a+kK)}$$

For $C(\infty) = 0.9$ **and** $T = 0.005$ **sec**

$$\frac{aK}{a+kK} = 0.9$$

$$10(a+kK) = \frac{1}{T^2} = 40000$$

Let $z = 1$**, then**

$$s^2 + (10+a)s + 10(a+kK)$$

$$= s^2 + 2\frac{z}{T}s + \frac{1}{T^2}$$

$$= s^2 + 400s + 40000$$

$K = 9.23$**,** $k = 391.12$ **and** $a = 390$

Lag compensator is

$$G_c(s) = \frac{k}{s+a} = \frac{391}{s+390} = \frac{1.00287}{1+0.002564s}$$

à **A feedback lag compensator is equivalent to a cascade lead compensator.**